Metradica

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Chapter 1

About

to describe

Part I

Part one: general introduction

Chapter 2

Background

- Tropical ecosystems, Amazon basin, focus on French Guiana
- Threats : droughts
- Trait-based ecology to define species strategies; a mechanistic understanding with hydraulic traits

Chapter 3

Project introduction

My thesis is included in the METRADICA strategic project lead by Clement Stahl and Ghislain Vieilledent. **Mechanistic traits to predict shifts in tree species abundance and distribution with climate change in the Amazonian forest**.

The global objectives of Metradica are to estimate species vulnerability to climate change and predict shifts in species distribution from species traits.

It combines: current species distribution, hydrological indices, climatic predictions, and use a few key traits at the individual level in innovative models (joint species distribution model; JSDMs), in order to better understand the processes by which species interact with their environment.

There are 4 tasks :

1. environmental index + species hydrological affinity
2. species distribution + functional responses to env't
3. prediction of species distribution from traits using *four-corners* joint species distribution models
4. shifts in species distribution through the lens of demographic process: interaction between models

I am involved in task 2 : to acquire additional traits information in order to have an understanding of the environmental effect on intra- and inter- specific trait variability for morpho-anatomic and mechanistic traits at regional and local scales.

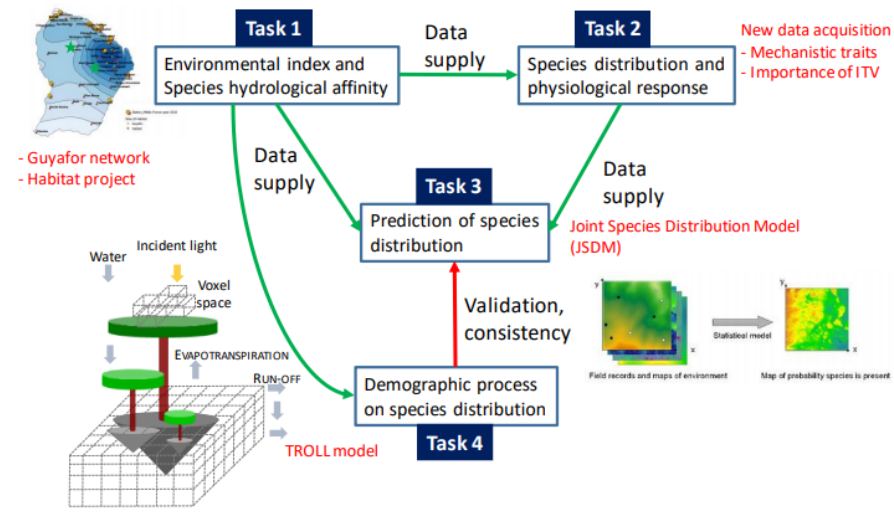


Figure 2: Operational organization of the project depicting the four tasks and their relationships

Figure 3.1: METRADICA

Chapter 4

Hypotheses

(1) Importance of local habitat in shaping functional strategies

- Habitats are widely recognized to control trait variation
- Identify species with statistically significant habitat associations
- Traits syndromes for specialists vs. generalists

I will focus on topographic-driven variation in the water table, which is hypothesized to explain variation in forest responses to drought (Nobre et al.,2011) and species distribution (Schietti et al., 2014; Lourenço Jr. et al., 2021), where trees with higher hydraulic efficiency and drought sensitivity are mostly found in seasonally flooded habitats (Oliveira et al., 2019), which seems to buffer the impact of short-term droughts on Amazon forest survival and productivity (Esteban et al., 2020; Sousa et al., 2020).

(2) Patterns of functional responses to drought along a precipitation and topographic gradient

- Explore how trait values vary across a precipitation gradient (Bafog "dry" to Kaw "wet")
- Are associations among traits consistent along the gradient in each habitat? Species should converge to a set of trait values that maximize their resource use efficiency. (Vleminckx et al 2021)
- Identify species in their limit range regarding drought resistance.

shifts in trait variation across environmental gradients can provide powerful insights into the drivers of community assembly (Junior et al 2021)

(3) Evaluating the inter- and intra-specific variation of hydraulic and leaf traits

Metradica data : exploring generalists species ITV (~60 individuals per species)
Hypothesis : because species' responses to the environment manifest through functional traits, the higher the ITV of a species is, the more diverse abiotic environments the species may be able to adapt to (Umaña et al., 2015).

Part II

Part two : Materials & Method

Chapter 5

Sampling

5.1 Habitats

Two main forest habitats are present in French Guiana, *terra firme* (high clay content and high water drainage) and *seasonally flooded forests* (nutrient rich soils and at least 3 months of annual flooding) (INSERER BIBLIO Ferry, 2010). The water table is never observed to descend below 60 cm depth and remains at the soil surface for at least two consecutive months each year (Baraloto et al., 2007; Ferry et al., 2010). During the field campaigns, we ensure that all trees sampled correspond to the correct habitat. In this case, any errors of habitat classification on the maps are being rectified.

The habitat classification of the tree species are based on all the past project achieved.

5.1.1 Hydrological indexes

Topographic variables are strong proxies for soil hydrology, which correlates with a combination of physico-chemical properties. A new hydrological index that accounts for both variations in climate and soil / topography across sites by combining the concept of MCWD with the one of Relative Extractable Water (e.g. Wagner et al. 2011) or Plant Available Water capacity (Nepstad et al. 2004, Ouédraogo et al. 2016). For Paracou, there is the topography classification established by Allié, Pélissier et al., 2015. *Sylvain*: The topographic wetness index ( ), identifies water accumulation areas. TWI was derived from 185 a 1-m resolution digital elevation model built based on data from a LiDAR campaign done in 186 2015 using SAGA-GIS (Conrad et al. 2015).

5.1.2 How do we define seasonally flooded soils?

On maps, pixels located at an altitude difference of less than 2 m from the altitude of the nearest surface run-off of the same catchment area. *pixels situés à moins de 2m de dénivelé du plus proche écoulement de surface appartenant au même bassin versant.* Surface run-off corresponds to pixels receiving the waterflow of at least 75 pixels upstream. Everything is being calculated from the SRTM 30 m (after adjusting the basin area with fillings.)

Gaelle information:

- utilisation couche sig onf
- critere pente (inf 20° = BF, pente moins forte espece de lissage avec bc d'arbres en BF, durcissement des criteres, affiner et bien tomber dans du BF) *comment:* SRTM pente 20° à 30m ne correspond pas n'ont plus à la pente sur le terrain, il y a une imprecision. but again in natura we make sure the tree is really in the corresponding habitat.
- distance à la crique (inf 50m) (marginale)

The classification in the waterlogged habitat may be loose and not as precise as for the *terra firme*, but it does not impact the sampling. As we mentioned above, *in natura* we assess if the tree is really in a waterlogegd habitat or not.

5.2 Sites

Permanent forest plots of the Guyafor network monitor since 1970 (Bafog, Paracou) individual tree growth and mortality in order to understand the different drivers of forest dynamics, including climate and disturbance. The network coves > 235 ha of tropical forest on 12 experimental sites. They are co-managed by the Cirad, CNRS and ONF institutes. METRADICA leaf samplings took place in the control plots.

Paracou : 9 of these permanent forest plots were subjected to 3 different forest exploitation treatments. Out of 9, 3 plots served as control plots. In 1992, 3 additional biodiversity / control plots of 6.25 ha in size and one 25 ha plot were established. Within these plots all trees with a diameter at breast height (DBH) above 10 cm were mapped. Species richness at Paracou ranges between 150 and 200 species per hectare. Since the establishment of the permanent forest plots, tree inventories take place at Paracou, recording tree status (alive or dead) and the DBH to the nearest 0.5 cm. Samplings were done in plots 6, 11, 13, 14, 15, and 16 from the 23/10/2020-7/12/2020 and from 13/09/2021- 17/09/2021. 244 trees sampled but 240 individuals with values. Lost 4 individuals (bota error usually)

Bafog: 5 permanent plots monitored of the ONF institute. Samplings were done in plots 2, 3, 4, and 5 and off plots from the 1st to 16th of March 2021.

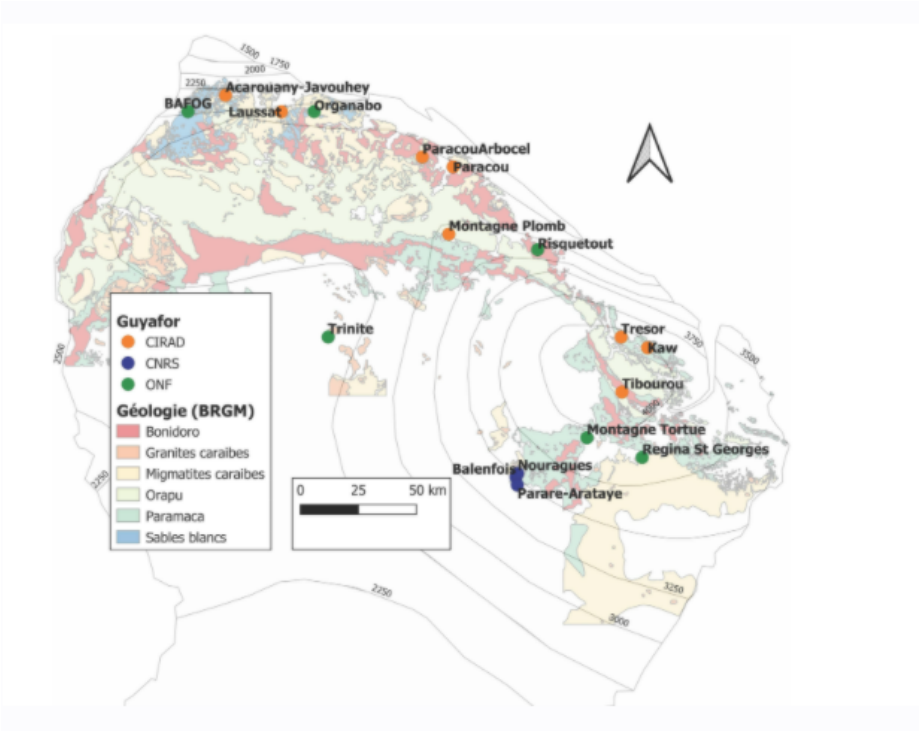


Figure 5.1: Guyafor stations for tropical forest monitoring

Est: Samplings were done in plots KawGuyafor (1 ha) and Trésor from the 4th to 23rd of October 2021. Some trees were also sampled outside the plots.

The three sites were chosen in order to sample species along the rainfall gradient.

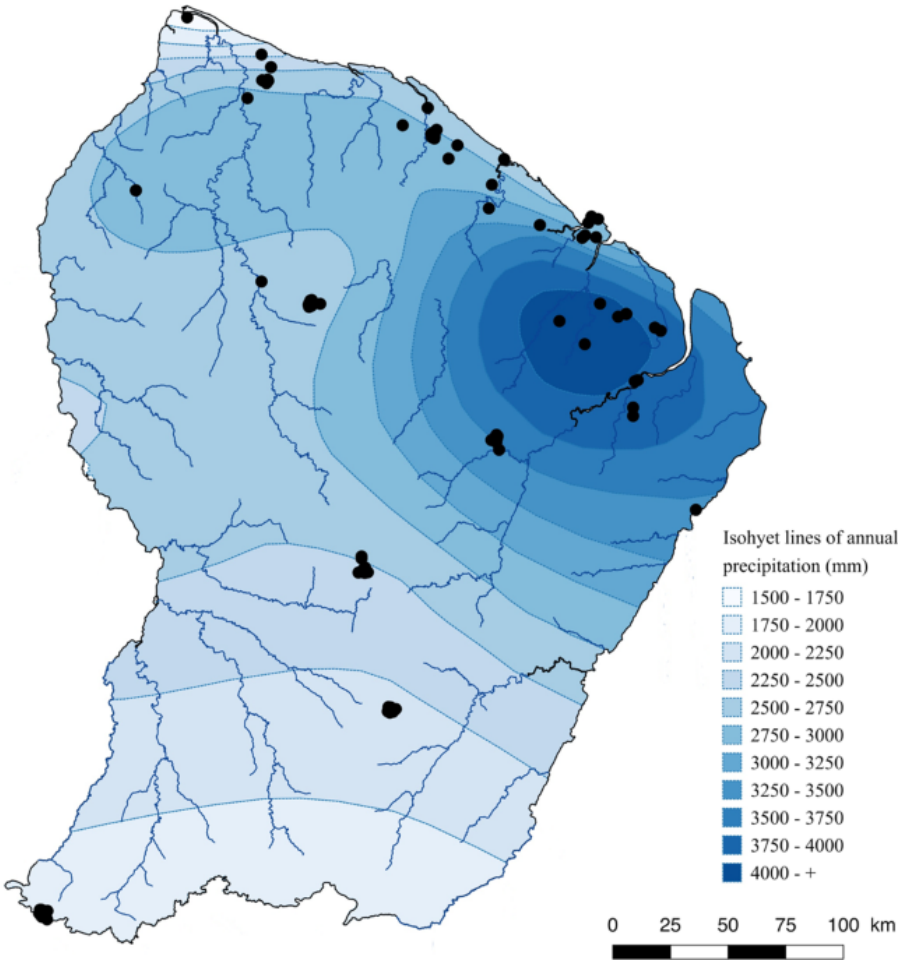


Figure 5.2: Rainfall gradient in French Guyana

5.3 Species

Specialist species are species that have a habitat preference, either for *terra firme* soils or for seasonally flooded soils in French Guiana. Generalist species are here defined as species able to thrive in the two contrasting habitats, they are regionally widespread and abundant species.

We **hypothesize** that they should display large ITV since they inhabit a large ecological spectrum. We are interested to see how ITV of functional traits is structured for these generalist species.

5.3.1 Indicator Species Analysis

We explored the degree of habitat preference using **Indicator Species Analysis** (Dufrene & Legendre 1997). It takes account of both relative abundance and relative frequencies of each species across the two main habitats in French Guiana, seasonally flooded forest and Terra firme forests.

- Generalist: IndVal 0 in the habitats (pVal>0.08)
- Specialists: IndVal > 0.24 in one habitat (pVal<0.08)

Species with high IndVals means that the species prefers the habitat (but not exclusive, hence the word **preference**), but are also a high probability of being sampled in the given habitat.

Our analysis identified 9 generalist taxa, 8 SFF specialist taxa and 7 terra firme specialist taxa.

5.3.2 More details about IndVal

IndVal used by other projects (ex. INSERER BIBLIO TerSteege et al 2013). The relative abundance and the relative frequency must be combined by multiplication because they represent independent information about the species distribution. X100 for a percentage. The index is max when all individuals of a species are found in a single group of sites or when the species occurs in all sites of that group. The statistical significance of the species indicator values is evaluated using a randominization procedure. Significance of habitat association was estimated by a Monte Carlo procedure that reassigns species densities and frequencies to habitats 1000 times. It therefore gives an ecological meaning as it compares the typologies.

Other indicators:

- Contrary to **TWINSPAN**, this indicator index for a given species is independent of other species relative abundances and there is no need for pseudospecies
- Species richness (sensitive to several factors)
- Oliviera species association to the topographic and edaphic gradient : weighted the HAND or P concentration of the plot by the abundance of the species in the plot. Divided by the number of individuals of the species in all plots.

5.3.3 Chosen species

Our analysis identified 9 generalist taxa, 8 SFF specialist taxa and 7 terra firme specialist taxa.

Family	Genus	Species	Type	Paracou	FTH	ParacouT
Fabaceae	Bocoa	prouacensis	Generalists	15	NA	
Sapotaceae	Chrysophyllum	prieurii	Generalists	12	5	
Euphorbiaceae	Conceveiba	guianensis	Generalists	18	2	
Lecythidaceae	Eschweilera	coriacea	Generalists	20	NA	
Chrysobalanaceae	Hymenopus	heteromorphus	Generalists	20	NA	
Bignoniaceae	Jacaranda	copaia	Generalists	18	2	
Burseraceae	Protium	stevensonii	Generalists	4	8	
Fabaceae	Tachigali	melinonii	Generalists	15	1	
Myristicaceae	Virola	michelii	Generalists	6	5	
Meliaceae	Carapa	surinamensis	SFF specialists	10	NA	
Fabaceae	Eperua	falcata	SFF specialists	11	NA	
Myristicaceae	Iryanthera	hostmannii	SFF specialists	9	NA	
Salicacea	Laetia	procera	SFF specialists	8	1	
Burseraceae	Protium	opcaum	SFF specialists	10	NA	
Fabaceae	Pterocarpus	officinalis	SFF specialists	10	NA	
Clusiaceae	Symphonia	globulifera	SFF specialists	10	NA	
Myristicaceae	Virola	surinamensis	SFF specialists	1	5	
Salicacea	Casearia	javitensis	TF specialists	6	5	
Fabaceae	Dicorynia	guianensis	TF specialists	6	NA	
Lecythidaceae	Gustavia	hexapetala	TF specialists	4	1	
Myristicaceae	Iryanthera	sagotiana	TF specialists	5	NA	
Chrysobalanaceae	Licania	membranacea	TF specialists	5	NA	
Metteniusaceae	Poraqueiba	guianensis	TF specialists	11	NA	
Fabaceae	Vouacapoua	americana	TF specialists	5	NA	

Un espèce : bc mortlité chez les jeunes donc parti pris de les considerer dans le choix des habitats. (percentile 10-90). For *Eperua falcata*: BF?? see Laurens porter 2007 (are species adapted …)

5.3.4 Phylogeny

We also wanted to maximize the phylogeny to get a greater picture, having specialists and generalists in the main clades: (Rosids, Asterids, Magnoliids). This would allow us to assess whether the different strategies to drought tolerance can be extend to the species of the same clade.

Table 5.1: Studied species.

Genus	Species
Eperua	falcata
Iryanthera	hostmannii
Jacaranda	copaia subsp. copaia
Pterocarpus	officinalis
Symphonia	globulifera
Tachigali	melinonii
Virola	surinamensis
Bocoa	prouacensis
Chrysophyllum	prieurii
Conceveiba	guianensis
Eschweilera	coriacea
Hymenopus	heteromorphus
Protium	stevensonii
Virola	michelii
Carapa	surinamensis
Laetia	procera
Protium	opacum
Casearia	javitensis
Dicorynia	guianensis
Gustavia	hexapetala
Iryanthera	sagotiana
Licania	membranacea
Poraqueiba	guianensis
Vouacapoua	americana

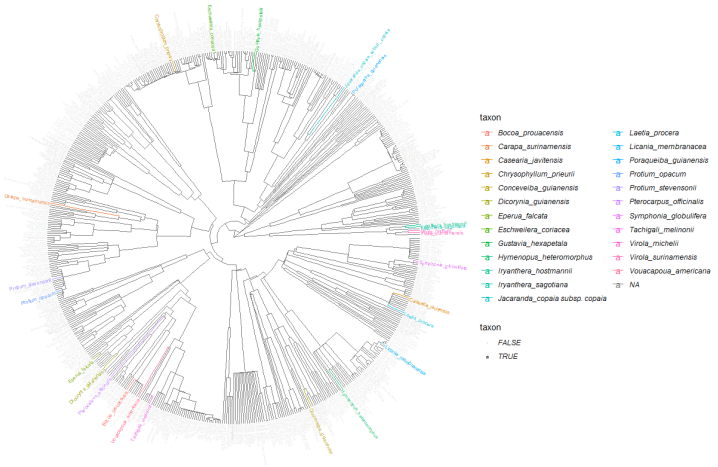


Figure 5.3: Phylogeny

5.4 Individuals

The individuals were selected with a DBH value between the 10th and the 90th percentile of the species' distribution pattern, in order to have a sampling that best reflects the forest structure. In this percentile interval, species were randomly selected for sampling.

Sampling strategy :

- Maps were produced by QGIS software
- **Field** protocol
- Fill the fieldworksheet **Fieldworksheet**

☐ Assess tree and branch height

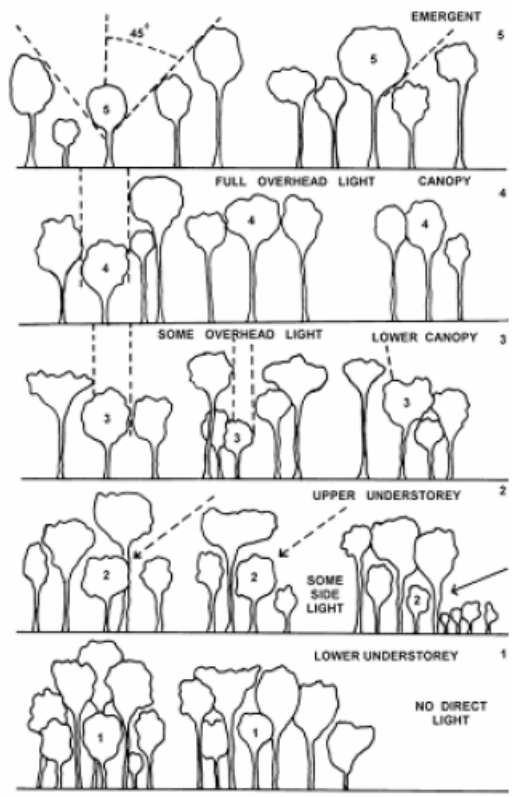


Figure 4-1. Indice de Dawkins (in Alder and Synnott 1992).

☐ Assess tree and branch Dawkins :

5.5 Summary

- 2 habitats (Terra firme; Seasonally flooded soils)
- 24 species (9 generalists; 8 SFF; 7 TF)
- 10 individuals per species per site
- 9 traits to investigate for drought tolerance

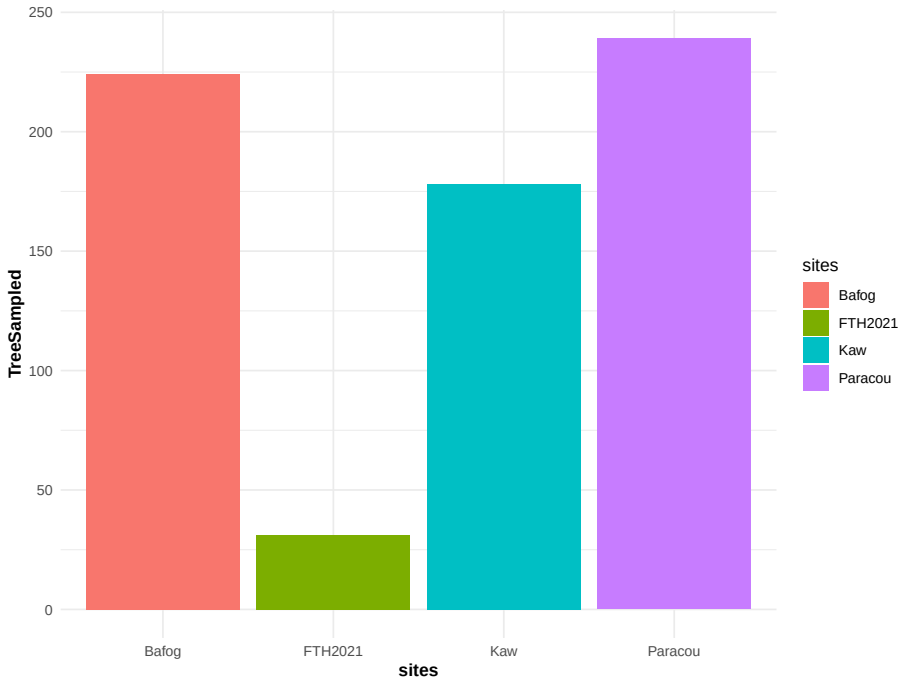

250
200
150
100
50
0
TreeSampled
Bafog
FTH2021
Kaw
Paracou
sites
sites
Bafog
FTH2021
Kaw
Paracou

Table 5.2: Individuals per species collected on the 3 sites

Family	Genus	Species	Type	Paracou	FTH	ParacouTotal	Bafog
Fabaceae	Bocoa	prouacensis	Generalists	15	NA	15	10
Sapotaceae	Chrysophyllum	prieurii	Generalists	12	5	17	11
Euphorbiaceae	Conceveiba	guianensis	Generalists	18	2	20	16
Lecythidaceae	Eschweilera	coriacea	Generalists	20	NA	20	19
Chrysobalanaceae	Hymenopus	heteromorphus	Generalists	20	NA	20	14
Bignoniaceae	Jacaranda	copaia	Generalists	18	2	20	12
Burseraceae	Protium	stevensonii	Generalists	4	8	12	11
Fabaceae	Tachigali	melinonii	Generalists	15	1	16	13
Myristicaceae	Virola	michelii	Generalists	6	5	11	16
Meliaceae	Carapa	surinamensis	SFF specialists	10	NA	10	1
Fabaceae	Eperua	falcata	SFF specialists	11	NA	11	0
Myristicaceae	Iryanthera	hostmannii	SFF specialists	9	NA	9	9
Salicacea	Laetia	procera	SFF specialists	8	1	9	10
Burseraceae	Protium	opcaum	SFF specialists	10	NA	10	0
Fabaceae	Pterocarpus	officinalis	SFF specialists	10	NA	10	10
Clusiaceae	Symphonia	globulifera	SFF specialists	10	NA	10	9
Myristicaceae	Virola	surinamensis	SFF specialists	1	5	6	10
Salicacea	Casearia	javitensis	TF specialists	6	5	11	9
Fabaceae	Dicorynia	guianensis	TF specialists	6	NA	6	7
Lecythidaceae	Gustavia	hexapetala	TF specialists	4	1	5	9
Myristicaceae	Iryanthera	sagotiana	TF specialists	5	NA	5	5
Chrysobalanaceae	Licania	membranacea	TF specialists	5	NA	5	4
Metteniusaceae	Poraqueiba	guianensis	TF specialists	11	NA	11	10
Fabaceae	Vouacapoua	americana	TF specialists	5	NA	5	7

Chapter 6

Traits

This chapter defines measured and sampled traits with associated protocols.

6.1 Morphological traits

- **LA**: leaf area cm^2 : carbo- acquisition related trait which informs on the light capture strategy.
- **SLA**: leaf specific area $cm^2.g^-1$ (leaf area/ dry mass).

6.2 Hydraulic traits

- **LSWC** : Leaf-saturated water content (g g−1) was calculated as the mass of leaf water (WT − WD) divided by leaf dry mass (WD).
- π_{TLP}: leaf water potential at which leaf cells lose turgor (MPa), a key drought tolerance trait.
- g_{min}: leaf minimum conductance $mmol.m^{-2}.s^{-1}$ taking into account water losses through the cuticle and incompletely closed stomata. A key drought-avoidance trait.
- **Stomatal characteristics (size, density)**: morphological characteristic at the crossroads between water loss and the maximum rate of photosynthesis.(*INSERER BIBLIO Julio V. Schneider 2016*)
- **Nervation density**: determining the capacity of water supply in leaves. (*INSERER BIBLIO Julio V. Schneider 2016*)
- **Leaf chemistry** : To assess the leaves' concentration in carbon, nitrogen and phosphorous.
- **Pmidday**: Water potential at the driest point of the day.

- **δ 18 O and δ 13 C** : Informs patterns of carbon and water use, related to water use efficiency. (*INSERER BIBLIO Farquhar et al 1989*) Measured in Vienna.

6.3 Photochemical trait

- $\frac{Fv}{Fm}$: maximum quantum yield of photosystem II, which reflects photosynthesis efficiency ($\mu mol.m^{-2}.s^{-1}/\mu mol.m^{-2}.s^{-1}$). It is an indicator of photosynthetic performance, as it has been shown to be sensitive to drought constraints in tropical forests (*INSERER BIBLIO Manzi et al 2021*) but decline after other leaf hydraulic traits (*INSERER BIBLIO Trueba et al 2019*).

6.4 Measured traits

- Day 1
  - Fresh
    - On field: Pmidday
    - scan gmin leaves
    - fresh weight RWC
    - g_{min} weight

    - **Stomatal density** nail polish (3 replicates per indivual)
    - $\frac{Fv}{Fm}$
  - Day 2
    - saturated weight RWC
    - g_{min} finish weights (at least 2 points next day)
    - π_{TLP}
  - Day 5 (+72h)
    - Dry weight RWC
- Later on :
  - Stomatal observation and density counts
  - Measure of vein densities
  - Chemistry of dry-grinded leaves

6.5 Protocols

- **Field** protocol

- π_{TLP} protocol
- **RWC** protocol
- **Stomata** protocol
- **FvFm** protocol
- **gmin** protocol
- **leaf venation** protocol

Part III

Part three : Results

The data

Chapter 7

Hypotheses :

Part I (Habitat preference - type) : *specialists* species should have a more adequate set of trait to a given habitat than the generalists in the same habitat. Generalists would express a wider range of traits.

PArt II (Precipitation gradient) :

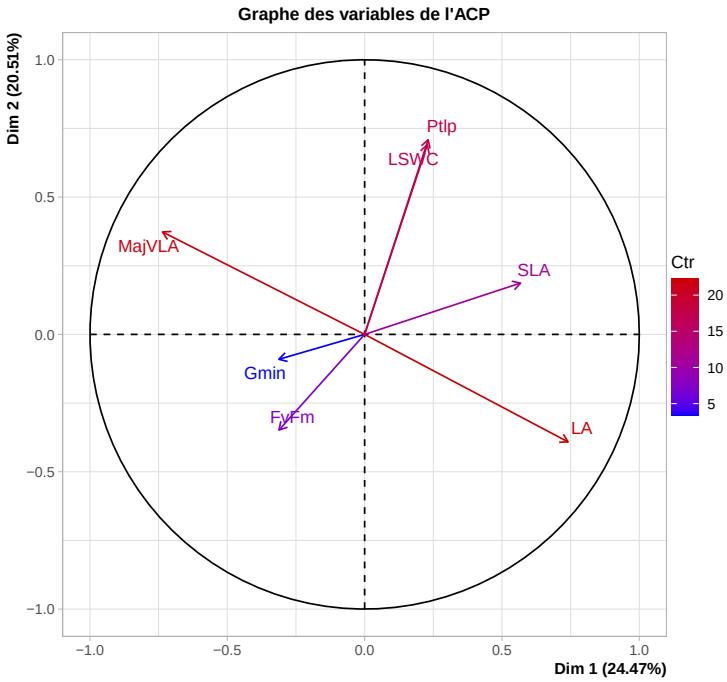
Chapter 8

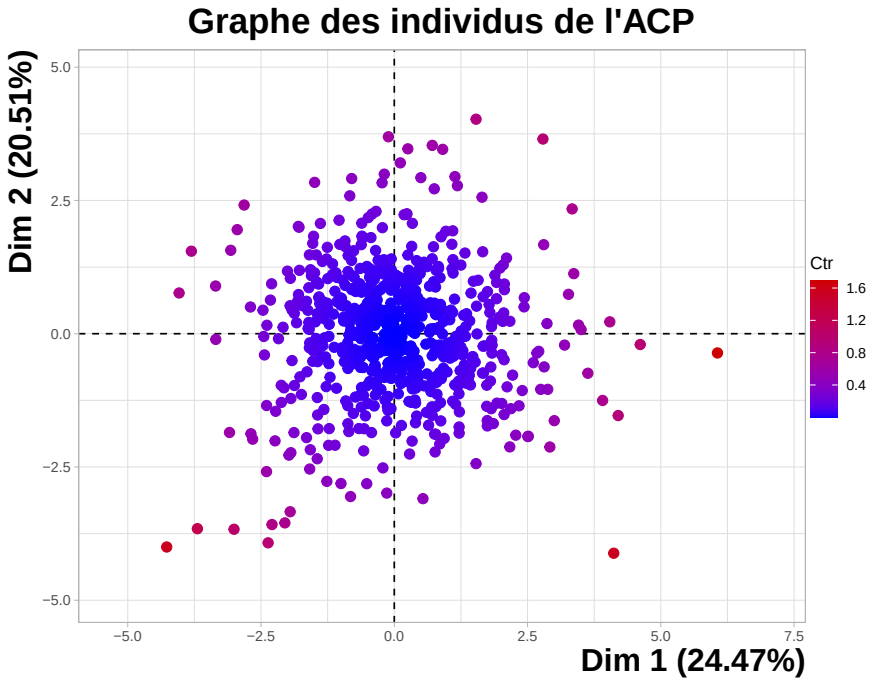
Data exploration

How to visualize the data and show the links between the variables and identify the same responses of the individuals?

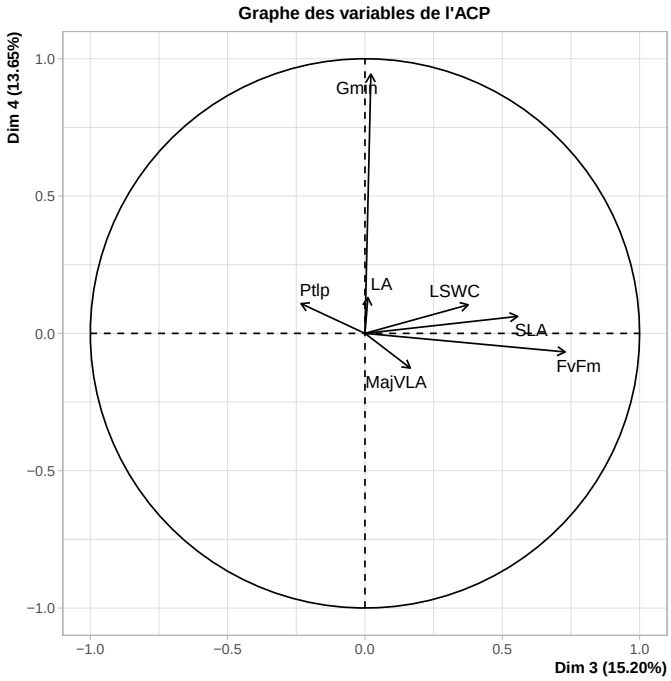
8.1 General PCA

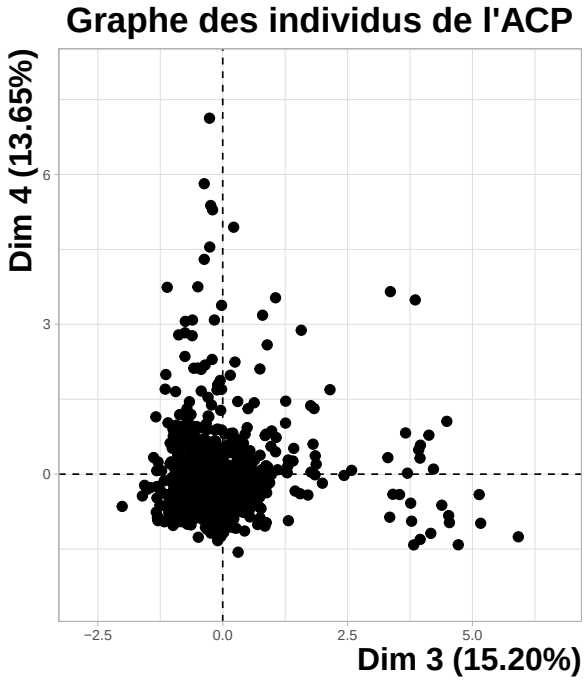
8.1.1 Contribution of individuals and variables : 1-2 dimensions



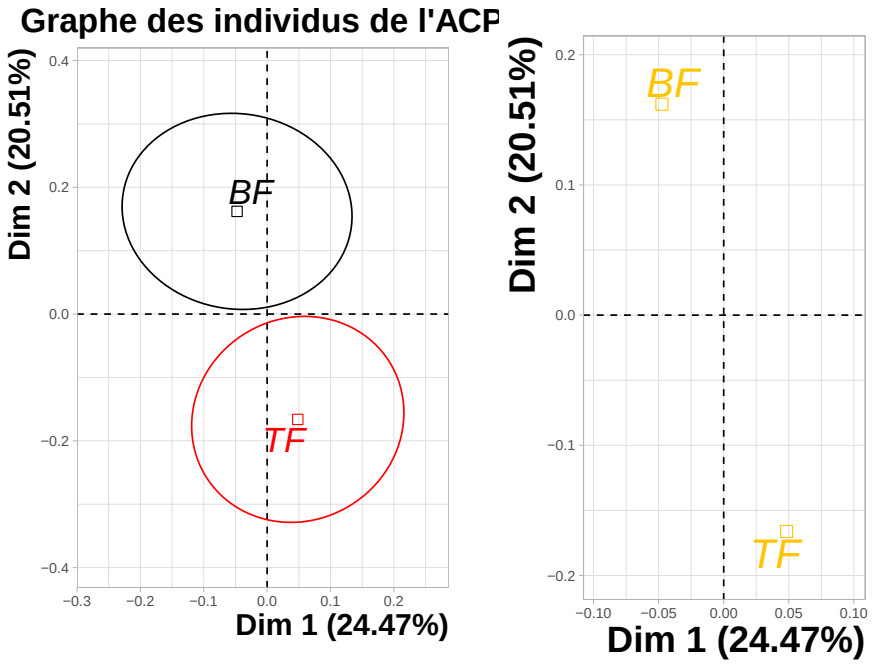


8.1.2 Contribution of individuals and variables : 1-2 dimensions

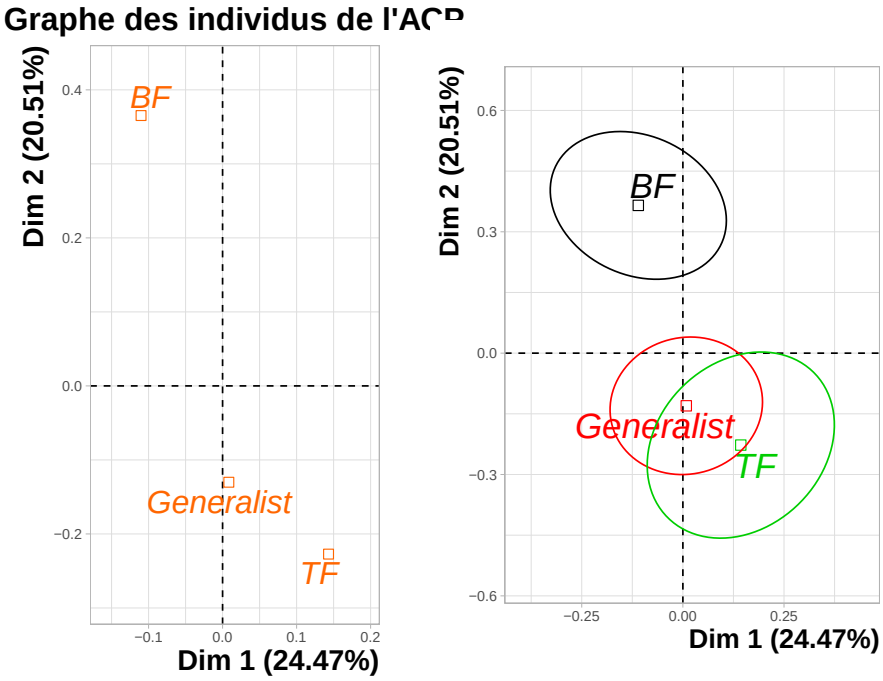




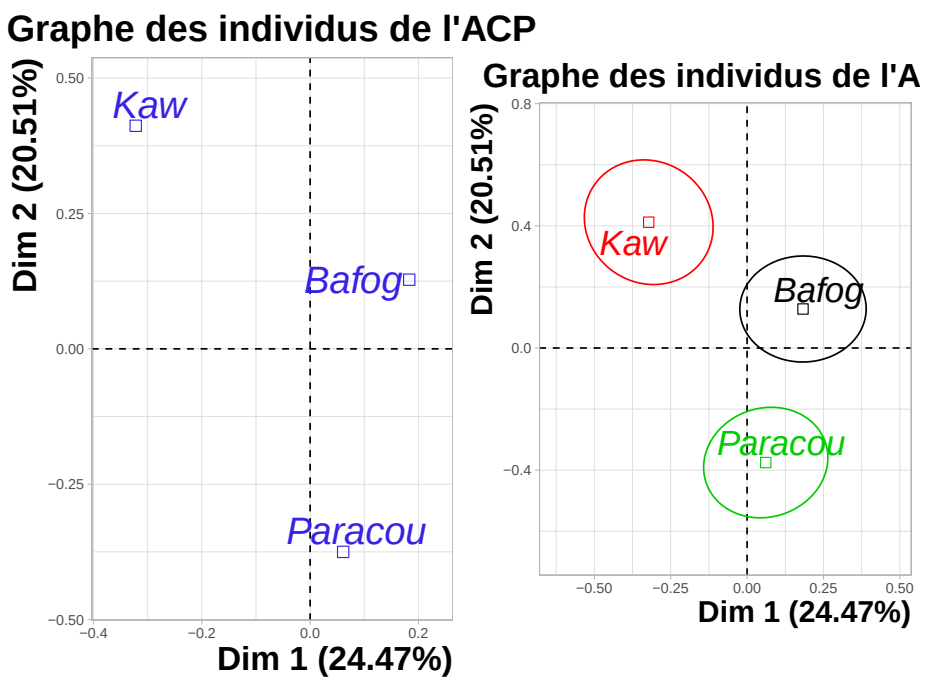
8.1.3 Habitat collected



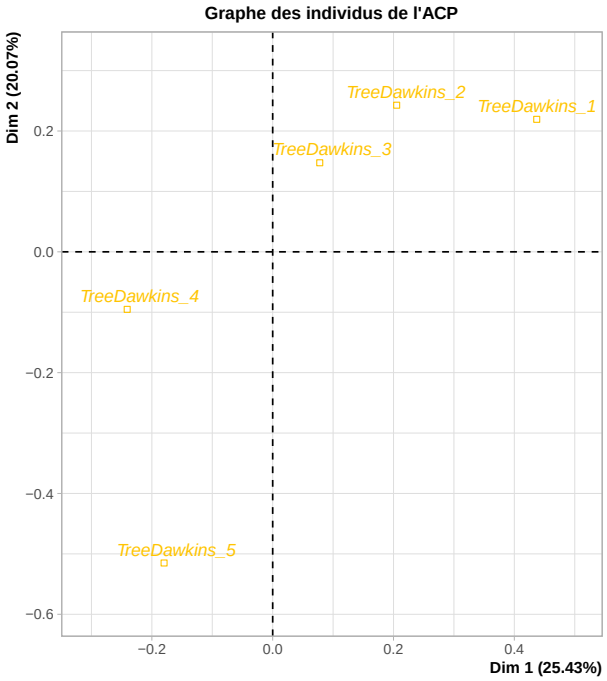
8.1.4 Habitat preference -type



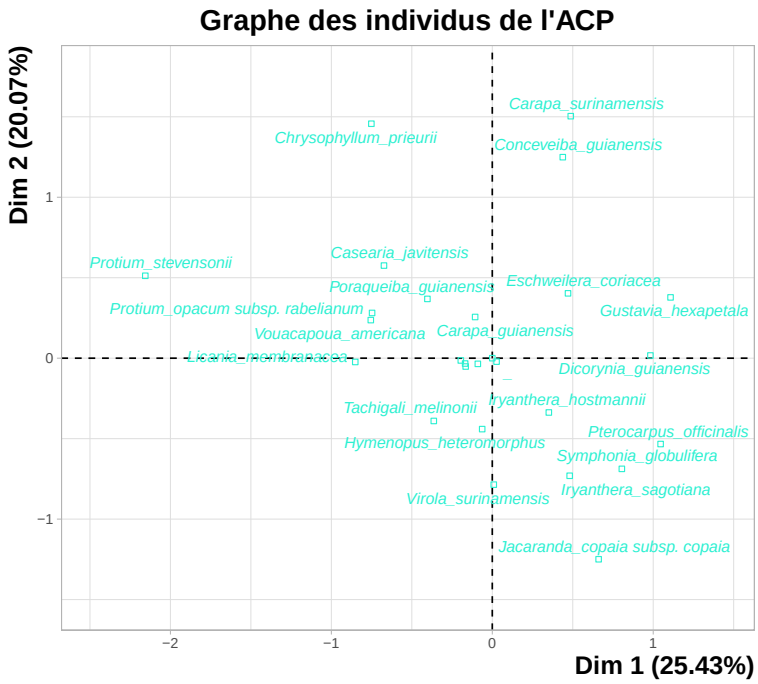
8.1.5 Forest sites



8.1.6 Dawkins of tree



8.1.7 Species



8.2 Distribution of major traits

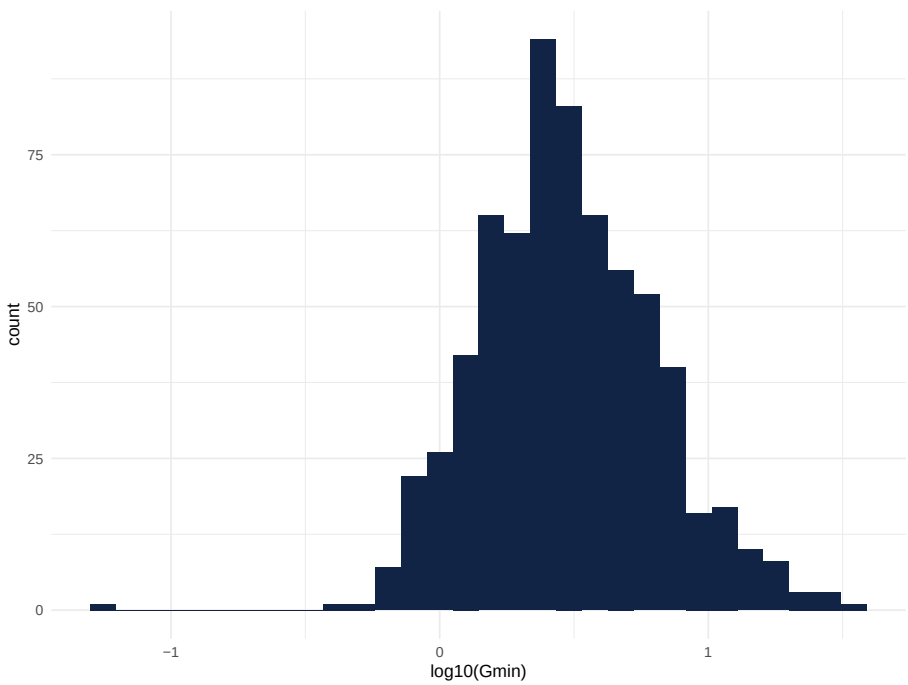
A focus on the frequency distribution of traits of individuals within communities allows scaling up from organism to ecosystem level and assessing how ecological communities and ecosystems respond to climate drivers (Garnier et al 2016; Liu et al 2020).

The shape of trait distribution could have the potential to reveal the ecological significance of trends and tradeoffs in traits at the species level. A focus on the *mean* and the *variance* of the community trait distribution is rooted in the concept of phenotype-environment matching and environmental optimality, where species' relative abundance is mediated by their traits, that is, "community assembly by trait selection". To understand how community trait composition is affected by ecological processes.

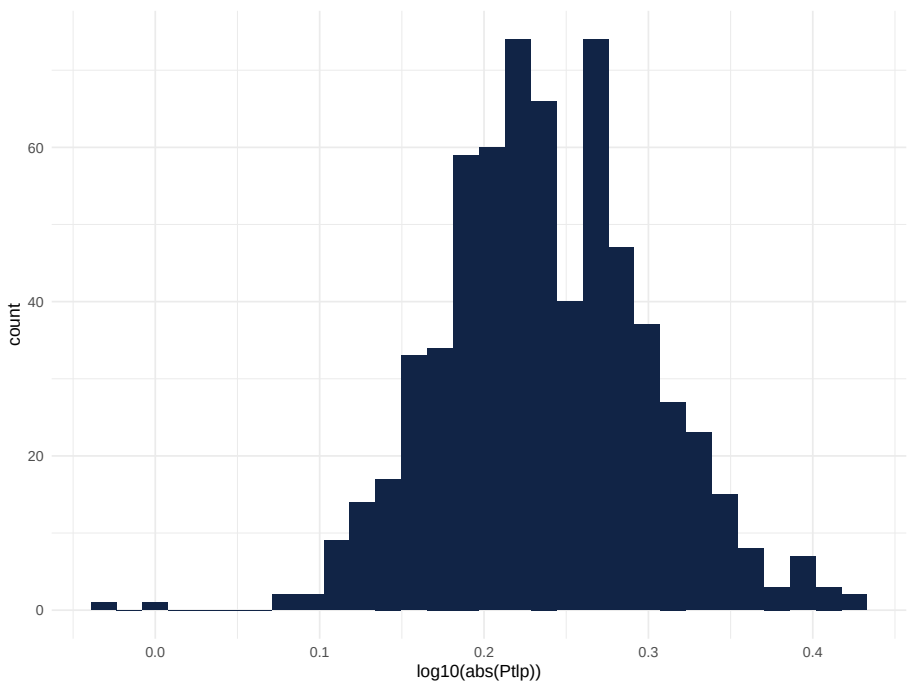
- use of community-weighted trait metrics CWM
- variance
- mean
- skewness (symmetry)
- kurtosis (measure of tailness with heavy tail referring to outliers)

after log tranform

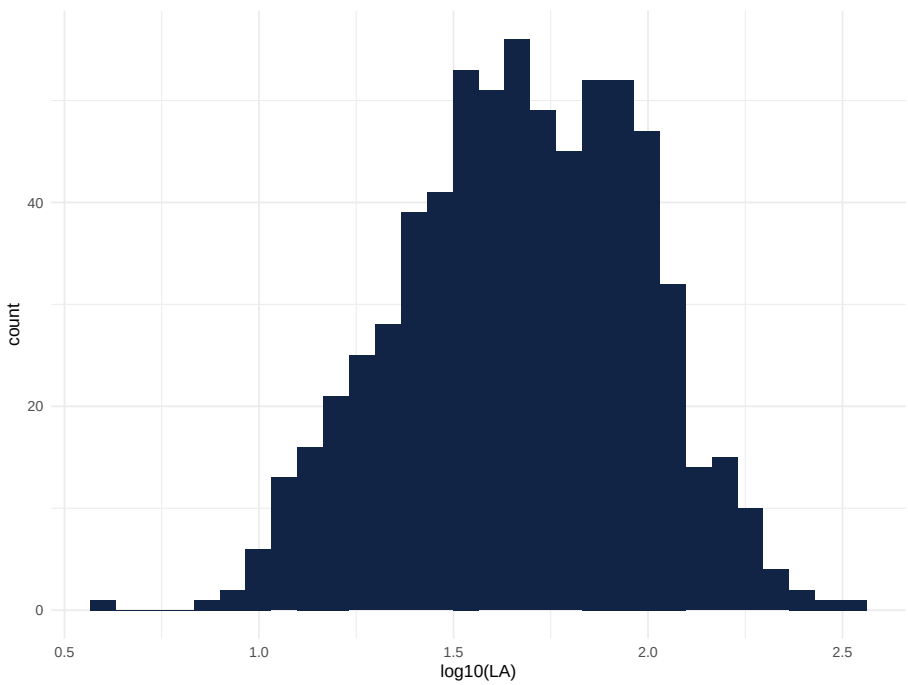
8.2.1 leaf minimal conductance



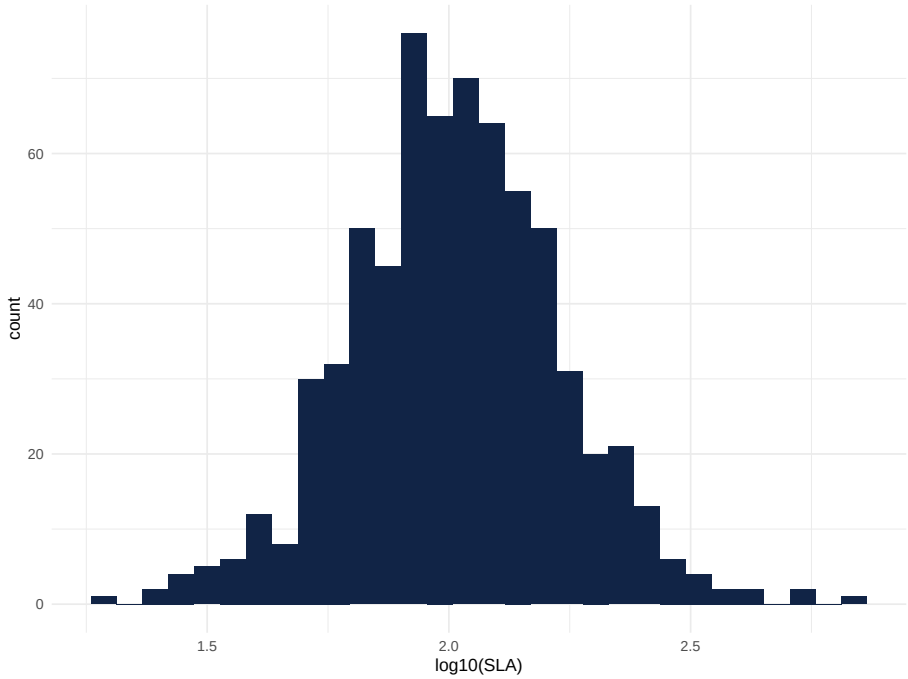
8.2.2 Turgor loss point



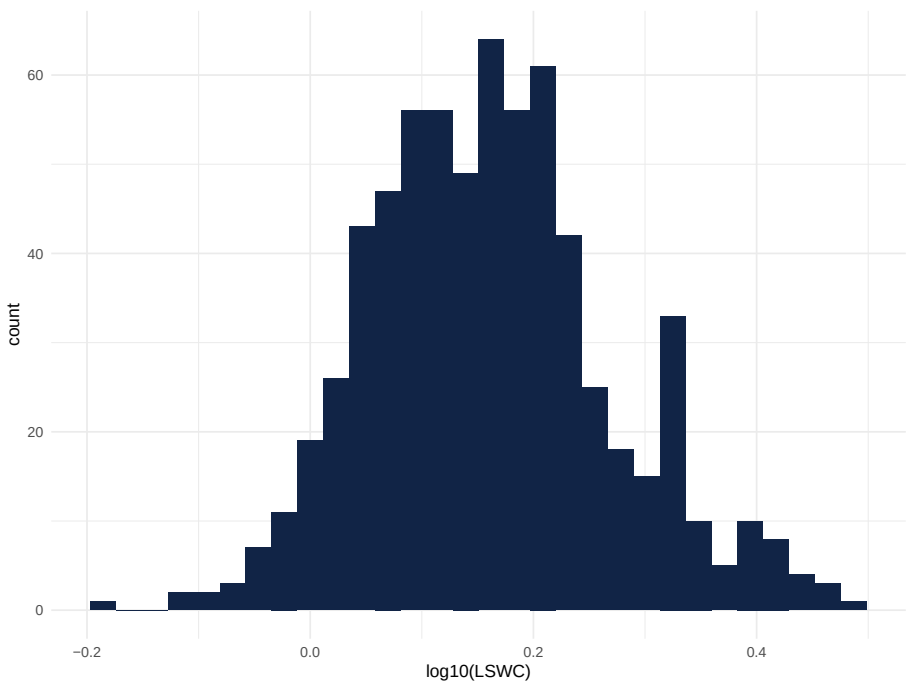
8.2.3 Leaf area



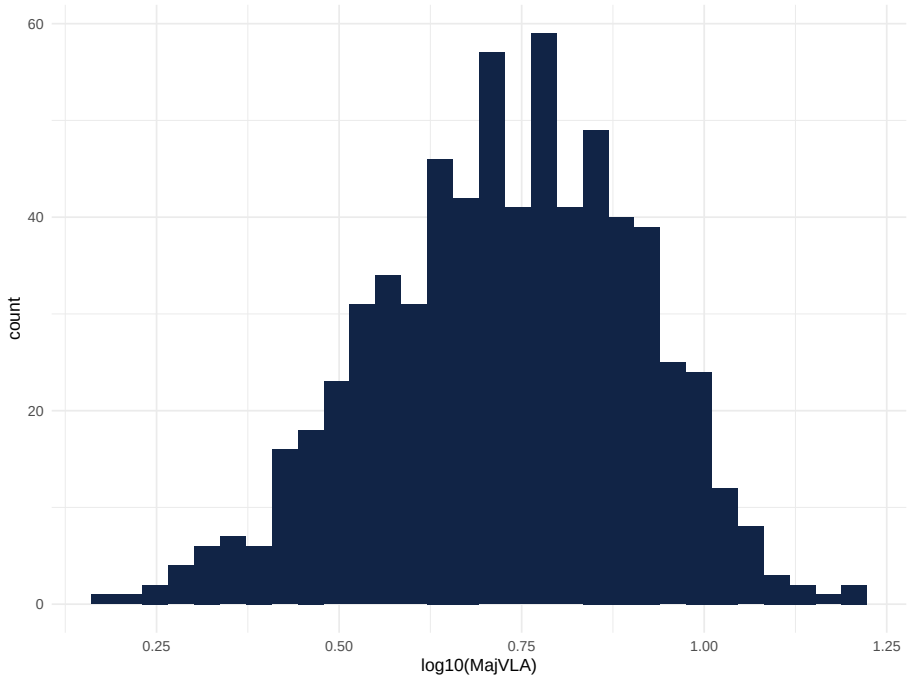
8.2.4 Specifig leaf area



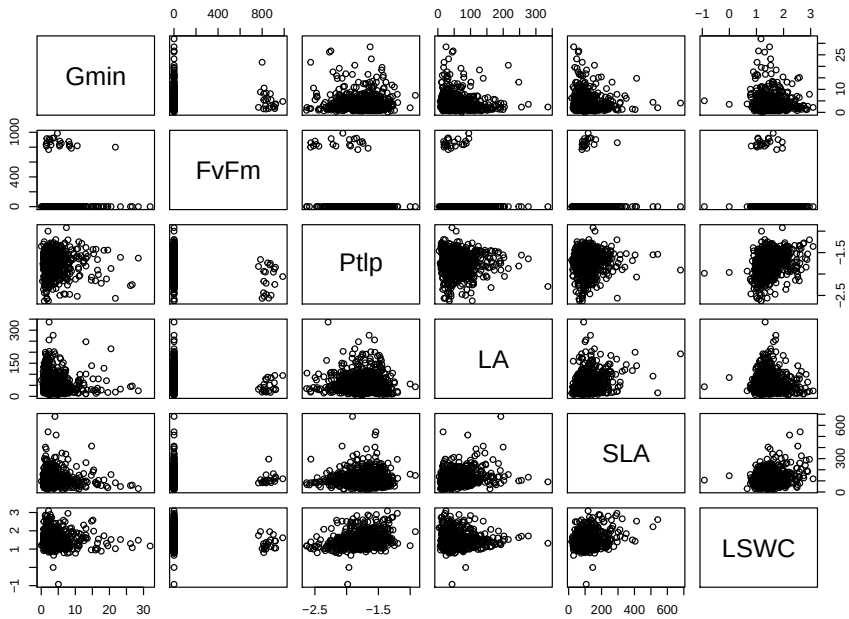
8.2.5 leaf saturated water content



8.2.6 Major vein density



8.3 Summary of main variables



8.4 Gmin - exploration

The capacity to retain water will become increasingly important for survival of tropical trees. Along with other plant hydraulic traits, cuticle conductance may strongly affect survival during droughts (Cochard, 2020).

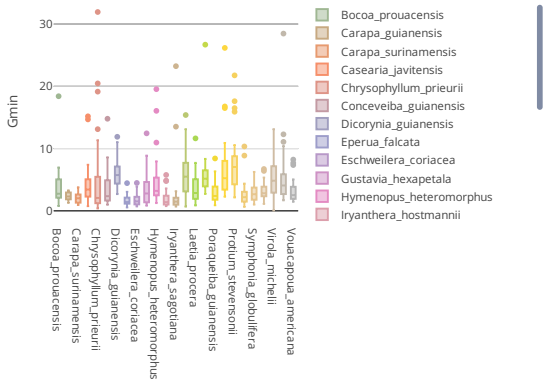
ANOVA reports a p-value far below 0.05, indicating there are differences in the means in the forest sites and habitat collect. Tukey post-hoc test shows groups a Bafog and b for Paracou and Kaw. It also shows groups TF a and BF b.

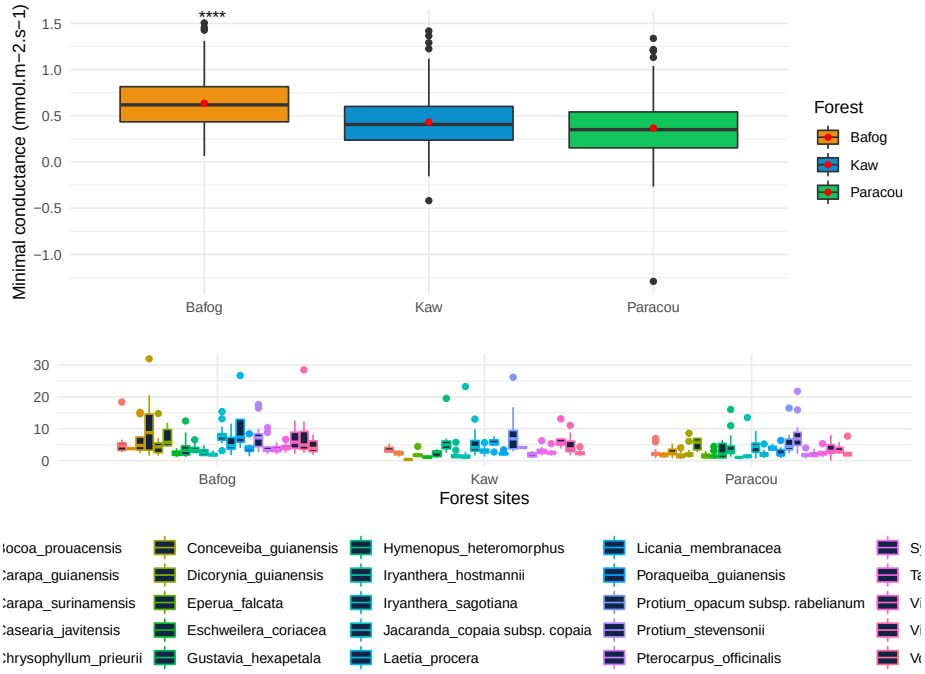
Model_gmin <- lm(log10(Gmin) ~ Forest + Habitat, data = Metradica)

```
## # A tibble: 3 x 2
##   Forest   Gmin
##   <chr>   <dbl>
## 1 Bafog    5.47
## 2 Kaw      3.60
## 3 Paracou  3.08
```

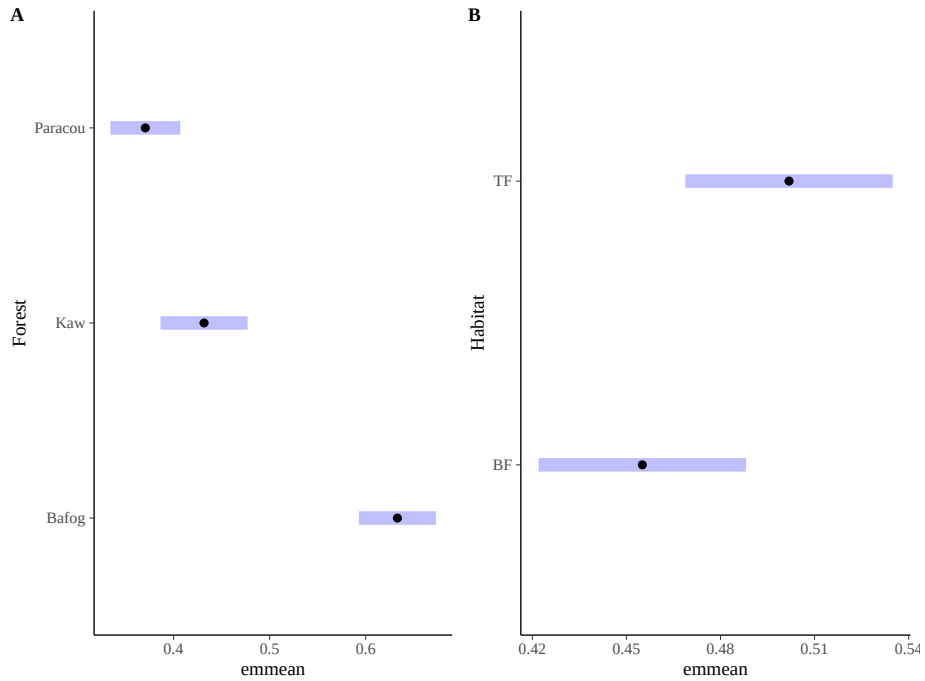
```
## Analysis of Variance Table
```

```
##
## Response: log10(Gmin)
##            Df Sum Sq Mean Sq F value  Pr(>F)
## Forest      2  9.396  4.6980 50.2291 < 2e-16 ***
## Habitat     1  0.364  0.3639  3.8911 0.04895 *
## Residuals 671 62.760  0.0935
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```





The adjusted mean arises when statistical averages must be corrected to compensate for data imbalances and large variances.

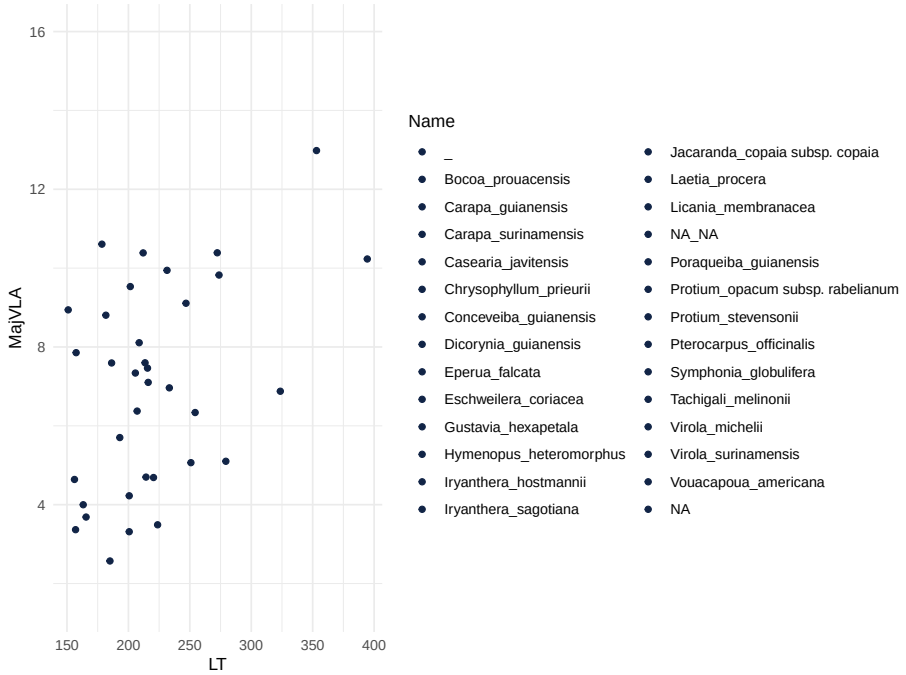


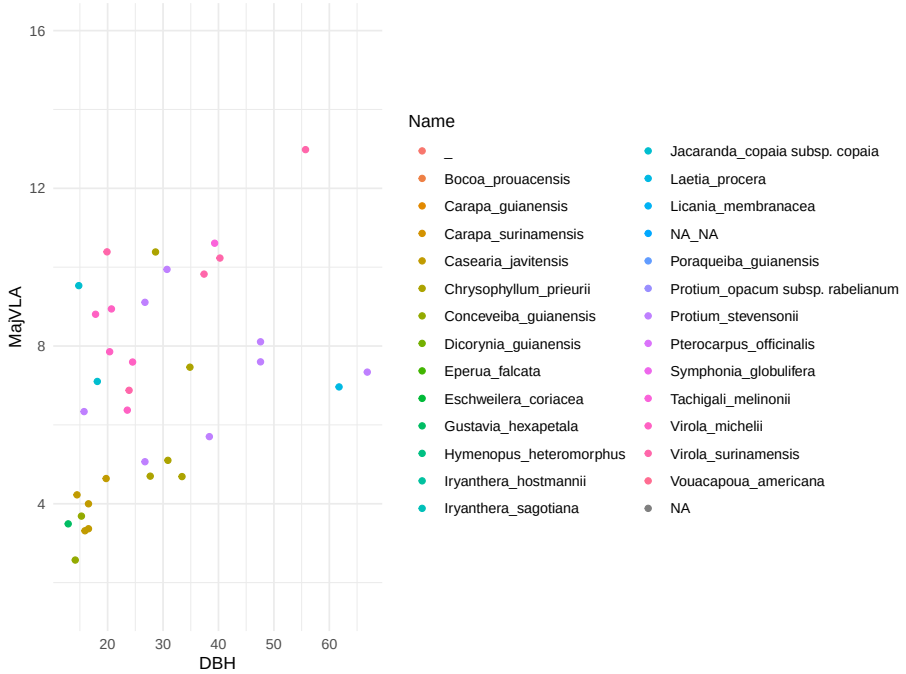
interpretation Across all species and all sites, gmin averaged 4.02 mmol.m-2.s-1. Gmin varied significantly among species (F-statistic: 18.10326, p-value: < 2.2204e-16). Across sites, gmin ranged from 3.08 mmol m–2 s–1 in Paracou to 3.60 mmol m–2 s–1 in Kaw to 5.47 mmol.m-2.m-1 i Bafog and site differences were significant (F-statistic: 50.0139, p-value: < 2.2204e-16).

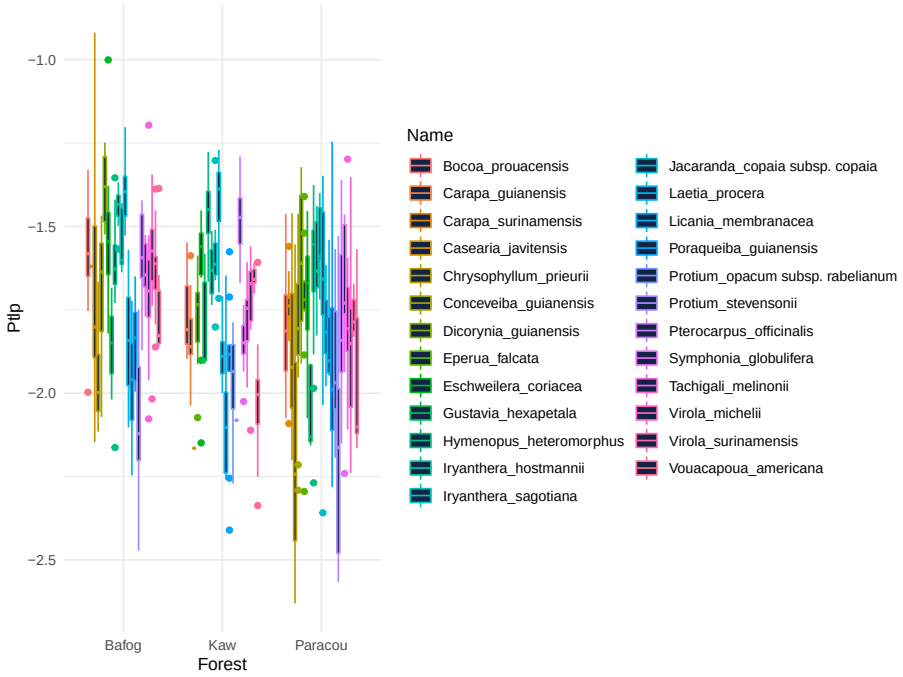
It is unclear whether a higher gmin in the Bafog site is assigned to stomatal opening or a higher permeability of the cuticle. We may need to consider soil water access as an additional factor to understand the pattern.

literature Slot et al 2021 found across 24 tropical species an average gmin of 4.0 mmol m–2 s–1. Large differences among species in cuticular water loss have the potential to contribute to differential mortality during drought, phenomena that are increasingly common in the tropics (Rifai et al 2019)

- **Test the phylogenetic signal.**
- **compare with stomatal density**
- **see if more trichomes for bafog site that could explain why these species can afford to have a higher gmin?**


16
12
8
4
MajVLA
150
200
250
300
350
400
LT
Name
_
Bocoa_prouacensis
Carapa_guianensis
Carapa_surinamensis
Casearia_javitensis
Chrysophyllum_prieurii
Conceveiba_guianensis
Dicorynia_guianensis
Eperua_falcata
Eschweilera_coriacea
Gustavia_hexapetala
Hymenopus_heteromorphus
Iryanthera_hostmannii
Iryanthera_sagotiana
Jacaranda_copaia subsp. copaia
Laetia_procera
Licania_membranacea
NA_NA
Poraqueiba_guianensis
Protium_opacum subsp. rabelianum
Protium_stevensonii
Pterocarpus_officinalis
Symphonia_globulifera
Tachigali_melinonii
Virola_michelii
Virola_surinamensis
Vouacapoua_americana
NA


16
12
8
4
MajVLA
20
30
40
50
60
DBH
Name
_
Bocoa_prouacensis
Carapa_guianensis
Carapa_surinamensis
Casearia_javitensis
Chrysophyllum_prieurii
Conceveiba_guianensis
Dicorynia_guianensis
Eperua_falcata
Eschweilera_coriacea
Gustavia_hexapetala
Hymenopus_heteromorphus
Iryanthera_hostmannii
Iryanthera_sagotiana
Jacaranda_copaia subsp. copaia
Laetia_procera
Licania_membranacea
NA_NA
Poraqueiba_guianensis
Protium_opacum subsp. rabelianum
Protium_stevensonii
Pterocarpus_officinalis
Symphonia_globulifera
Tachigali_melinonii
Virola_michelii
Virola_surinamensis
Vouacapoua_americana
NA


−1.0
−1.5
−2.0
−2.5
Ptlp
Bafog
Kaw
Paracou
Forest
Name
Bocoa_prouacensis
Carapa_guianensis
Carapa_surinamensis
Casearia_javitensis
Chrysophyllum_prieurii
Conceveiba_guianensis
Dicorynia_guianensis
Eperua_falcata
Eschweilera_coriacea
Gustavia_hexapetala
Hymenopus_heteromorphus
Iryanthera_hostmannii
Iryanthera_sagotiana
Jacaranda_copaia subsp. copaia
Laetia_procera
Licania_membranacea
Poraqueiba_guianensis
Protium_opacum subsp. rabelianum
Protium_stevensonii
Pterocarpus_officinalis
Symphonia_globulifera
Tachigali_melinonii
Virola_michelii
Virola_surinamensis
Vouacapoua_americana

Chapter 9

To do

What is missing: * stomatal densities * FTH data * vein densities * check gmin function * check for outliers -> see Daniela

Chapter 10

Import Paracou dataset

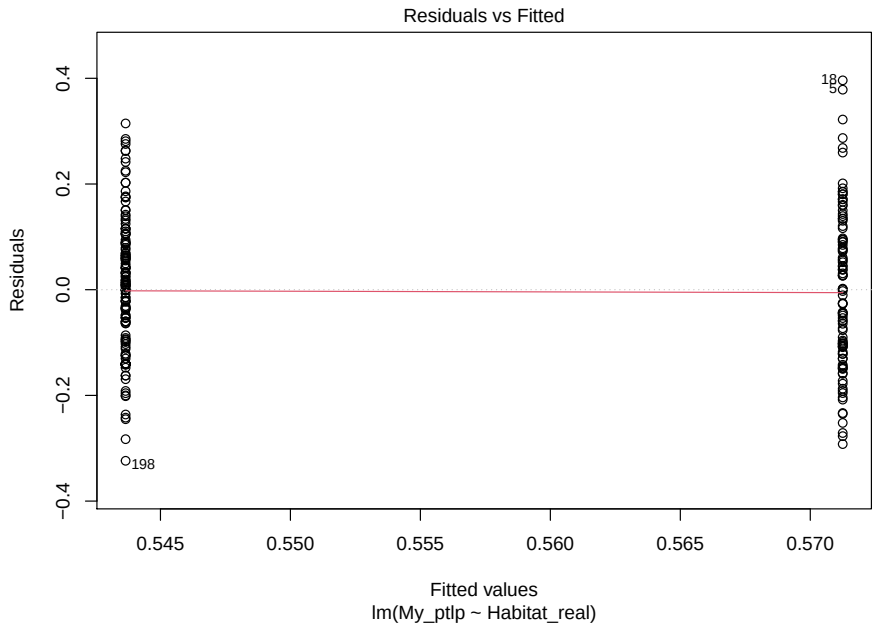
il manque encore les quelques individus résoltés avec Sylvain dans les indv de Paracou.

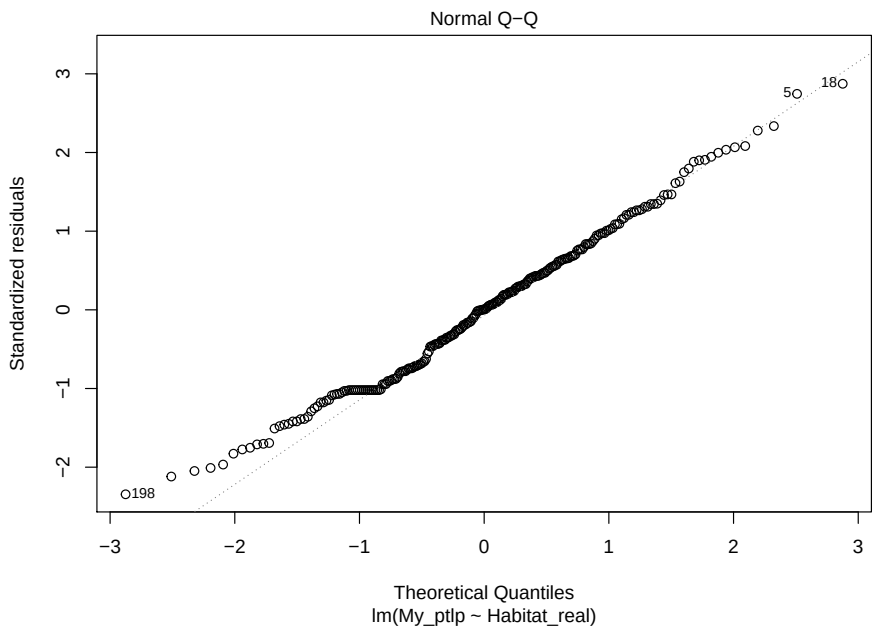
```
## # A tibble: 265 x 31
##    RWCnr_original idTree  Plot SubPlot TreeFieldNum SpecCode_CorrBota
##    <chr>           <dbl> <dbl>   <dbl>        <dbl> <chr>
##  1 M1             145493    15       1          182 Iry.hos
##  2 M2             147663    15       3          178 Gus.hex
##  3 M3             147783    15       3          298 Iry.hos
##  4 M4             147625    15       3          140 Tac.mel
##  5 M5             148586    15       3         1103 Chr.pri
##  6 M6             149072    15       4          369 Con.gui
##  7 M7             146131    15       1          824 Lic.mem
##  8 M8             149660    15       4          960 Iry.hos
##  9 M9             146342    15       2           84 Hym.het
## 10 M10            146466    15       2          209 Con.gui
## # ... with 255 more rows, and 25 more variables: spName_CorrBota.x <chr>,
## #   Genus_CorrBota <chr>, Species_CorrBota <chr>, Family_CorrBota <chr>,
## #   Tree_Dawkins <dbl>, Tree_Height <dbl>, Branch_Dawkins <dbl>,
## #   Branch_Height <dbl>, Branch_Position_crown <chr>, Habitat_preference <chr>,
## #   Habitat_real <chr>, ...1 <dbl>, spName_CorrBota.y <chr>,
## #   My_LSWC_E.per_corr <dbl>, My_RWC_E.per_corr <dbl>, My_ptlp <dbl>,
## #   My_posm <dbl>, My_gmin.slope <dbl>, My_gmin_manual <dbl>, ...
```

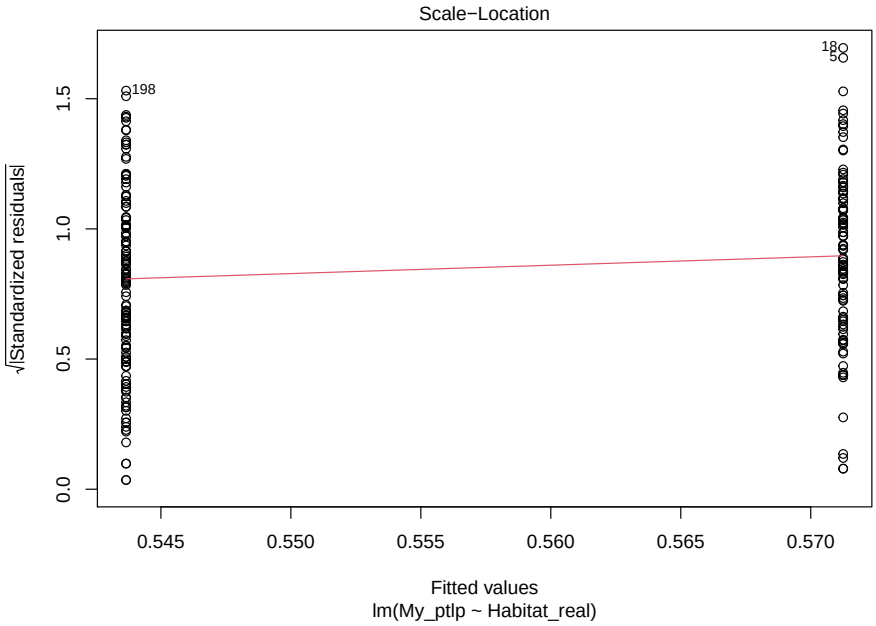
```
##
## Call:
## lm(formula = My_ptlp ~ Habitat_real, data = data)
##
## Residuals:
```

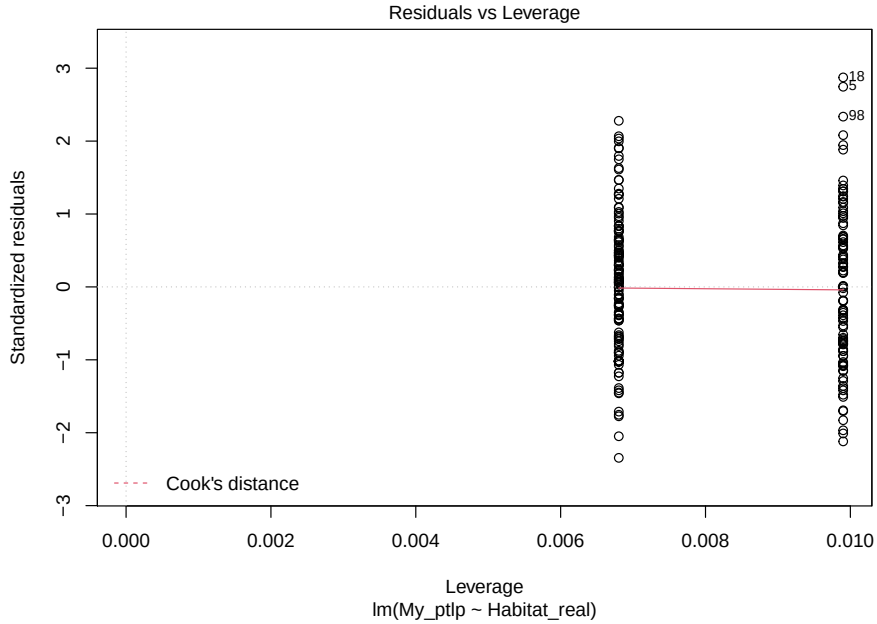
```
##         Min          1Q      Median          3Q         Max
## -0.32372154 -0.10954903  0.00086063  0.09043193  0.39611578
##
## Coefficients:
##                  Estimate Std. Error  t value Pr(>|t|)
## (Intercept)    0.54365996 0.01142657 47.57859  < 2e-16 ***
## Habitat_realTF 0.02758826 0.01790527  1.54079  0.12465
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1385398 on 246 degrees of freedom
##   (17 observations deleted due to missingness)
## Multiple R-squared:  0.009558298,    Adjusted R-squared:  0.005532112
## F-statistic: 2.374033 on 1 and 246 DF,  p-value: 0.1246533
```

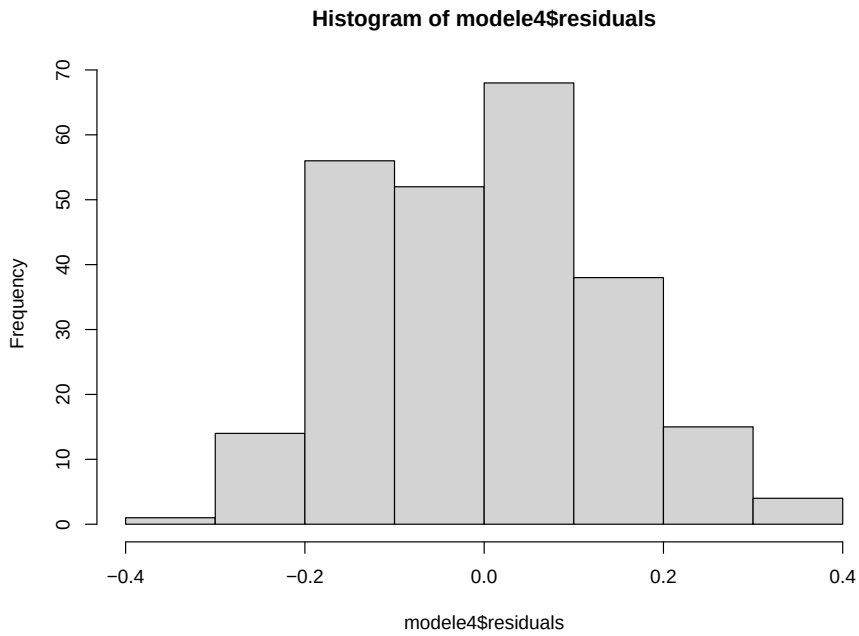
```
##
## Call:
## lm(formula = My_ptlp ~ Habitat_preference + Habitat_real, data = data)
##
## Residuals:
##         Min          1Q      Median          3Q         Max
## -0.33587496 -0.10881675 -0.01224824  0.08440152  0.42481904
##
## Coefficients:
##                                Estimate  Std. Error  t value Pr(>|t|)
## (Intercept)                  0.56957570  0.01636211 34.81066  < 2e-16 ***
## Habitat_preferenceGeneralist -0.04884120  0.02246211 -2.17438 0.030638 *
## Habitat_preferenceTF          0.02363455  0.03562569  0.66341 0.507692
## Habitat_realTF                0.02181046  0.02323052  0.93887 0.348726
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1359139 on 244 degrees of freedom
##   (17 observations deleted due to missingness)
## Multiple R-squared:  0.0544985,  Adjusted R-squared:  0.04287348
## F-statistic: 4.688035 on 3 and 244 DF,  p-value: 0.003340864
```

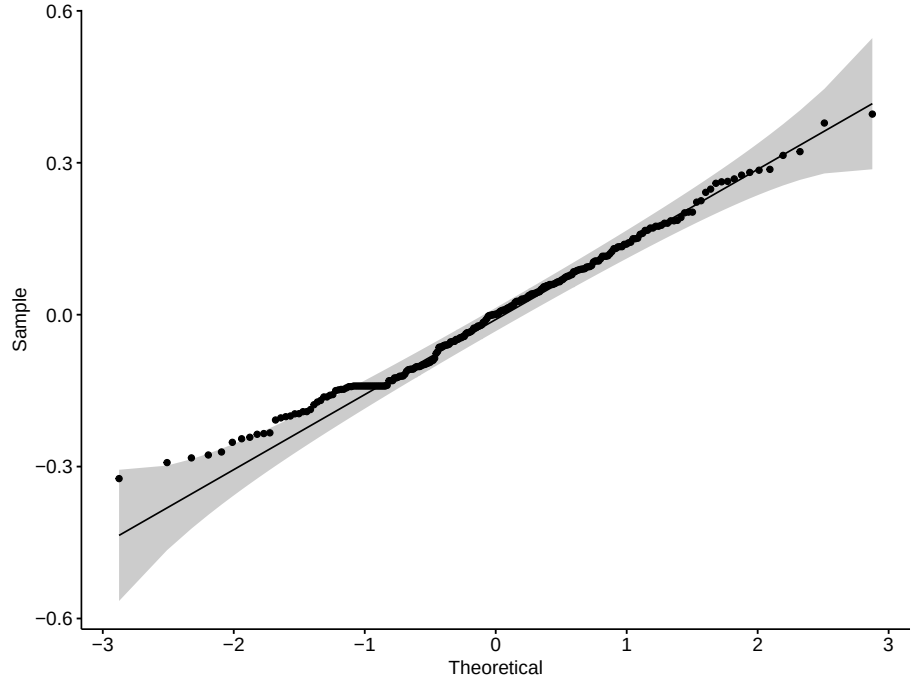

Residuals vs Fitted
Residuals
Fitted values
lm(My_ptlp ~ Habitat_real)
198
18
5
0.545
0.550
0.555
0.560
0.565
0.570
−0.4
−0.2
0.0
0.2
0.4


Normal Q−Q
Standardized residuals
Theoretical Quantiles
lm(My_ptlp ~ Habitat_real)
198
5
18
−3
−2
−1
0
1
2
3


Scale−Location
√|Standardized residuals|
Fitted values
lm(My_ptlp ~ Habitat_real)
198
18
5
0.0
0.5
1.0
1.5
0.545
0.550
0.555
0.560
0.565
0.570

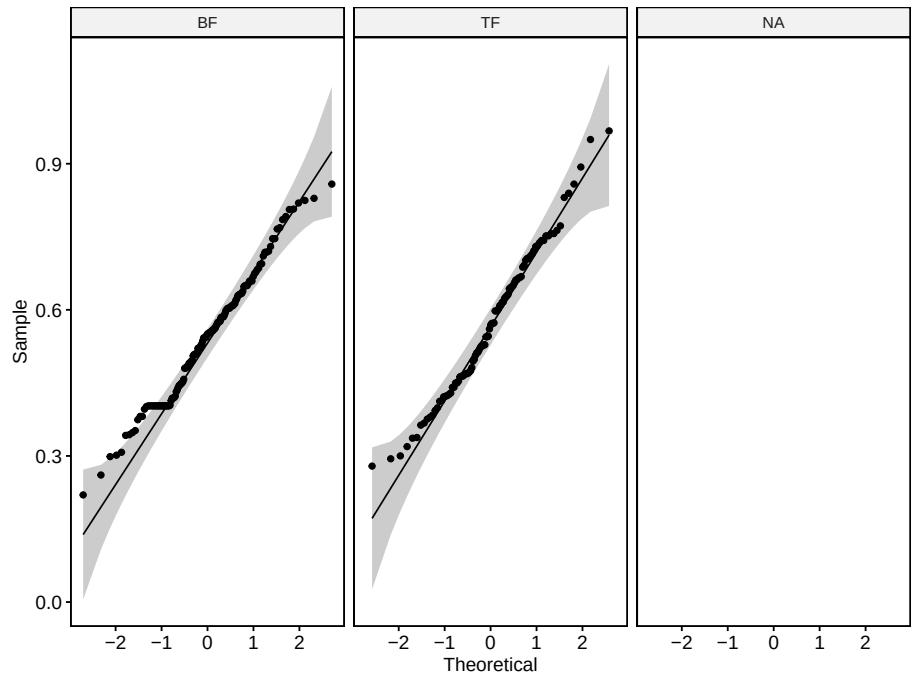

Residuals vs Leverage
Standardized residuals
Leverage
lm(My_ptlp ~ Habitat_real)
Cook's distance
18
5
98
−3
−2
−1
0
1
2
3
0.000
0.002
0.004
0.006
0.008
0.010


Histogram of modele4$residuals
Frequency
0
10
20
30
40
50
60
70
−0.4
−0.2
0.0
0.2
0.4
modele4$residuals


Sample
0.6
0.3
0.0
−0.3
−0.6
−3
−2
−1
0
1
2
3
Theoretical

```
## # A tibble: 1 x 3
```

```
##   variable           statistic p.value
##   <chr>                  <dbl>   <dbl>
## 1 residuals(modele4)     0.990  0.0961
```




```
## # A tibble: 1 x 4
##     df1   df2 statistic      p
##   <int> <int>     <dbl>  <dbl>
## 1     1   246      3.73 0.0546
```

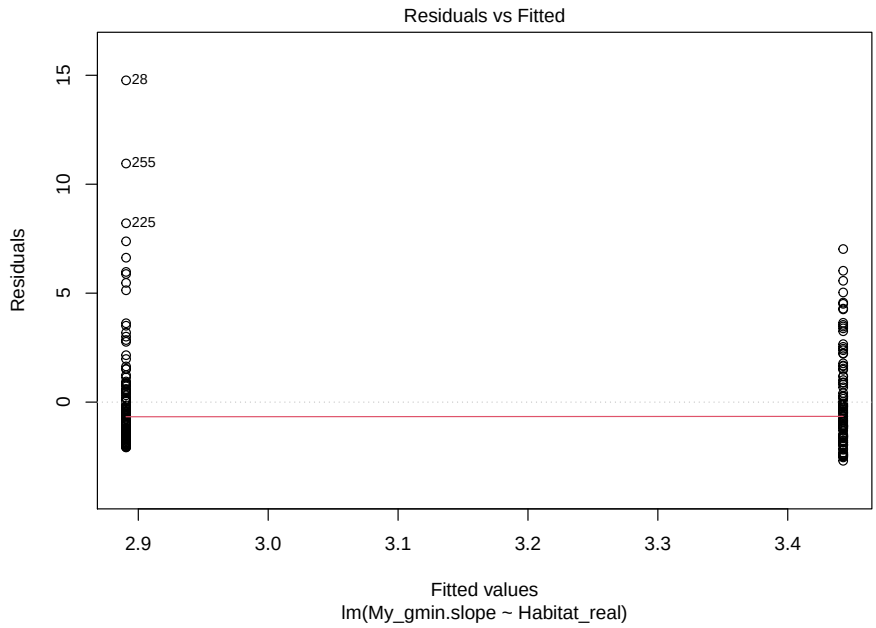
```
##
##  Bartlett test of homogeneity of variances
##
## data:  My_ptlp by Habitat_real
## Bartlett's K-squared = 2.507089, df = 1, p-value = 0.1133351
```

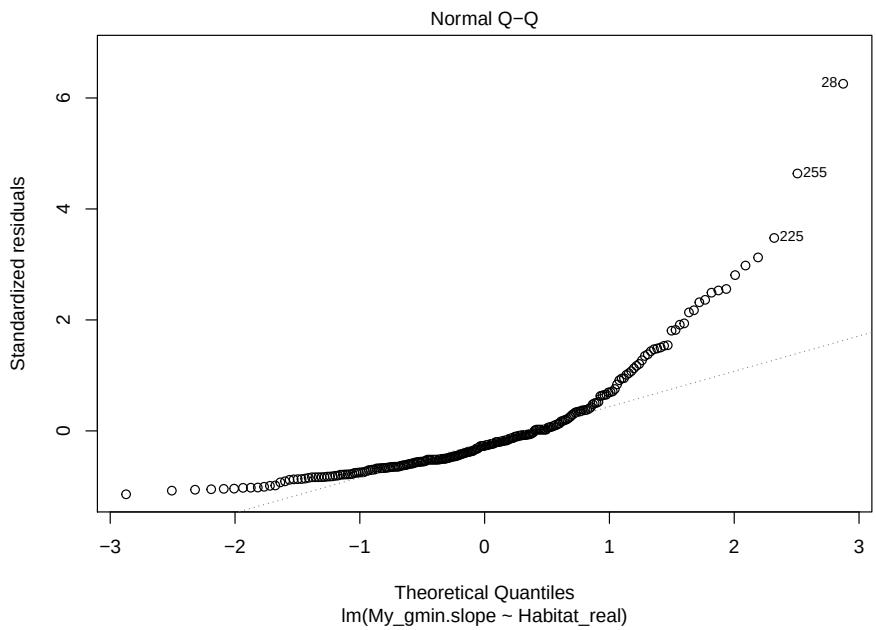
```
## # A tibble: 1 x 7
##   .y.         n statistic   DFn   DFd     p method
## * <chr>   <int>     <dbl> <dbl> <dbl> <dbl> <chr>
## 1 My_ptlp   265      2.25     1  194. 0.135 Welch ANOVA
```

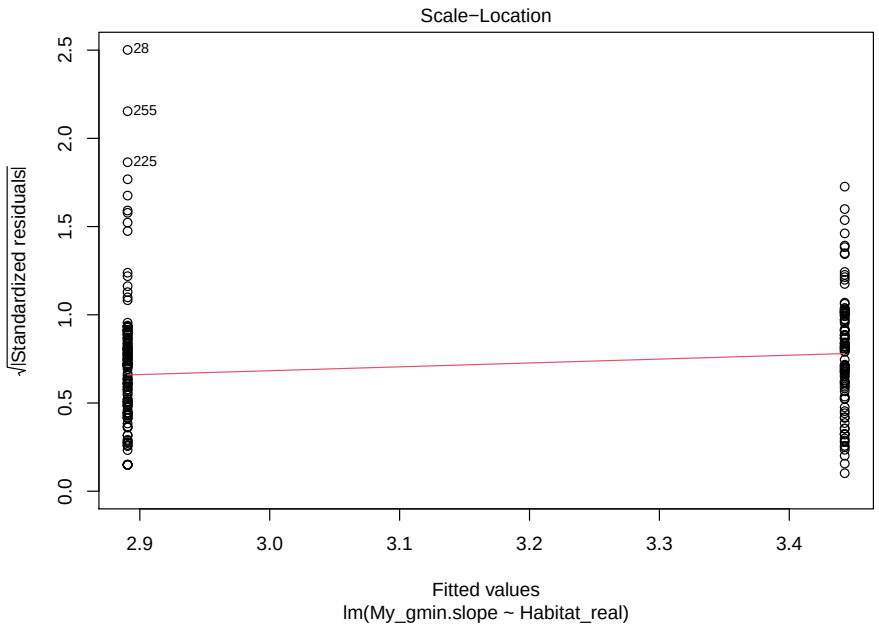
```
##
## Call:
```

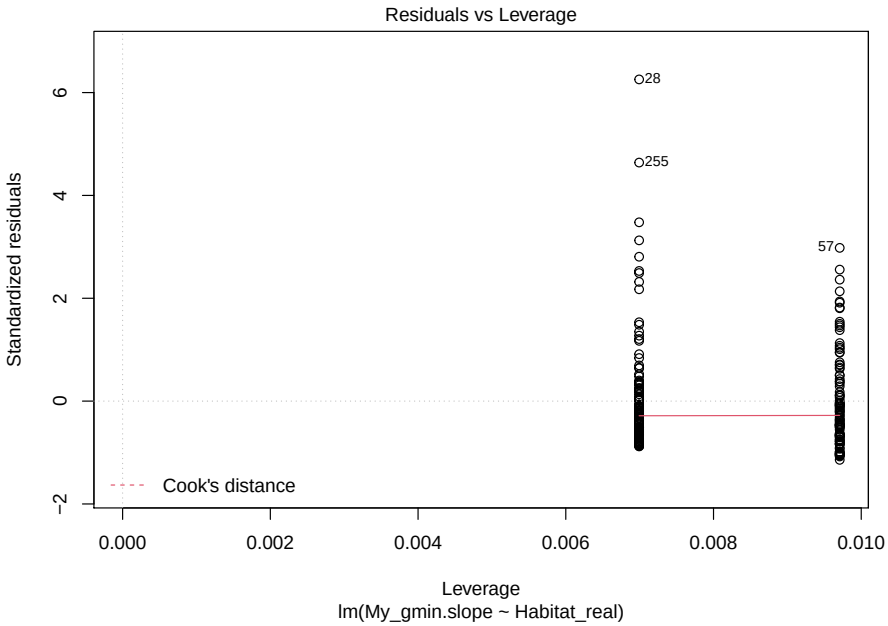
```
## lm(formula = My_gmin.slope ~ Habitat_real, data = data)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -2.6928590 -1.4901146 -0.6281806  0.5393048 14.7669882
##
## Coefficients:
##                 Estimate Std. Error  t value Pr(>|t|)
## (Intercept)    2.8905919  0.1980907 14.59227  < 2e-16 ***
## Habitat_realTF 0.5520879  0.3061350  1.80341 0.072557 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.36882 on 244 degrees of freedom
##   (19 observations deleted due to missingness)
## Multiple R-squared:  0.01315377, Adjusted R-squared:  0.009109313
## F-statistic: 3.252299 on 1 and 244 DF,  p-value: 0.07255689
```

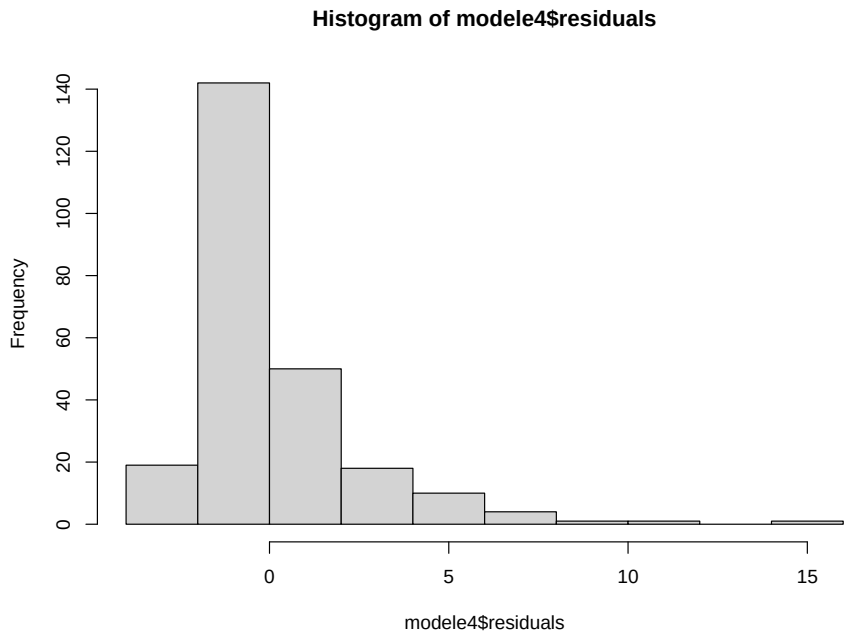
```
##
## Call:
## lm(formula = My_gmin.slope ~ Habitat_preference + Habitat_real,
##     data = data)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -2.7767711 -1.5100875 -0.6961231  0.6157774 14.3335811
##
## Coefficients:
##                               Estimate Std. Error t value   Pr(>|t|)
## (Intercept)                 2.39896597 0.28728977 8.35034 5.2913e-15 ***
## Habitat_preferenceGeneralist 0.92503300 0.39407727 2.34734   0.019715 *
## Habitat_preferenceTF        1.06222272 0.61751070 1.72017   0.086680 .
## Habitat_realTF              0.06540324 0.40067098 0.16323   0.870470
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.351568 on 242 degrees of freedom
##   (19 observations deleted due to missingness)
## Multiple R-squared:  0.03544718, Adjusted R-squared:  0.02348991
## F-statistic: 2.964488 on 3 and 242 DF,  p-value: 0.03277794
```

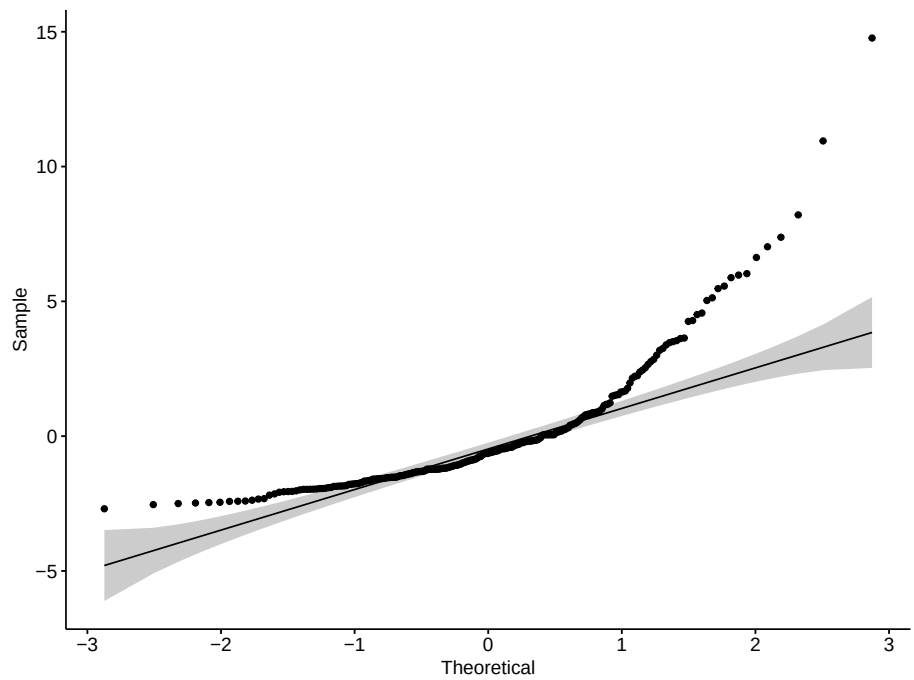

Residuals vs Fitted
Residuals
Fitted values
lm(My_gmin.slope ~ Habitat_real)
28
255
225
15
10
5
0
2.9
3.0
3.1
3.2
3.3
3.4


Normal Q−Q
Standardized residuals
Theoretical Quantiles
lm(My_gmin.slope ~ Habitat_real)
28
255
225
6
4
2
0
−3
−2
−1
0
1
2
3


Scale−Location
√|Standardized residuals|
28
255
225
0.0
0.5
1.0
1.5
2.0
2.5
2.9
3.0
3.1
3.2
3.3
3.4
Fitted values
lm(My_gmin.slope ~ Habitat_real)

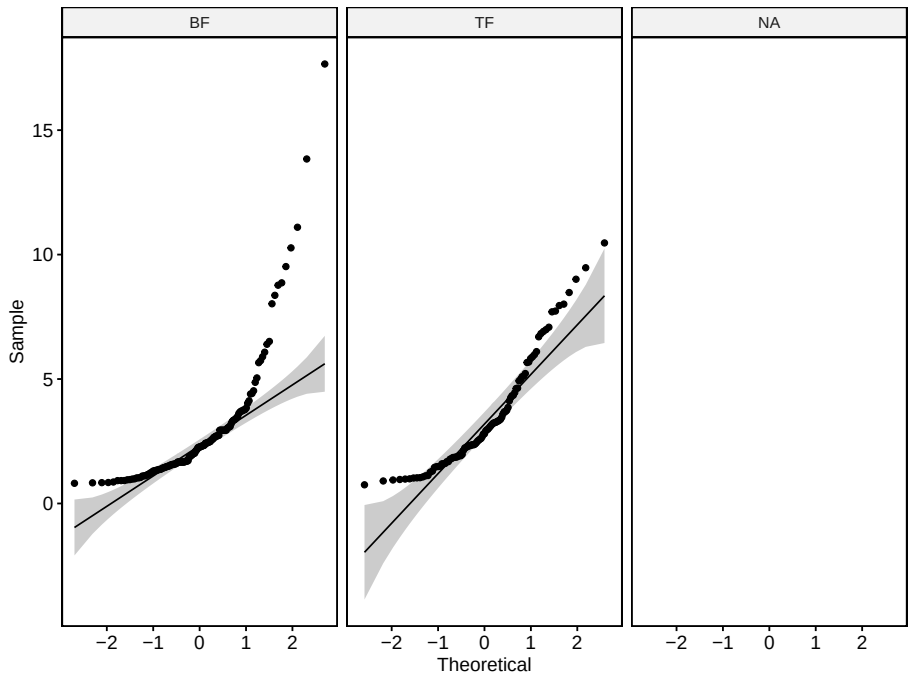

Residuals vs Leverage
Standardized residuals
28
255
57
Cook's distance
−2
0
2
4
6
0.000
0.002
0.004
0.006
0.008
0.010
Leverage
lm(My_gmin.slope ~ Habitat_real)


Histogram of modele4$residuals
Frequency
0
20
40
60
80
100
120
140
0
5
10
15
modele4$residuals


Sample
15
10
5
0
−5
−3
−2
−1
0
1
2
3
Theoretical

```
## # A tibble: 1 x 3
```

```
##   variable           statistic  p.value
##   <chr>                  <dbl>    <dbl>
## 1 residuals(modele4)     0.779 6.28e-18
```




```
## # A tibble: 1 x 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     1   244     0.878 0.350
```

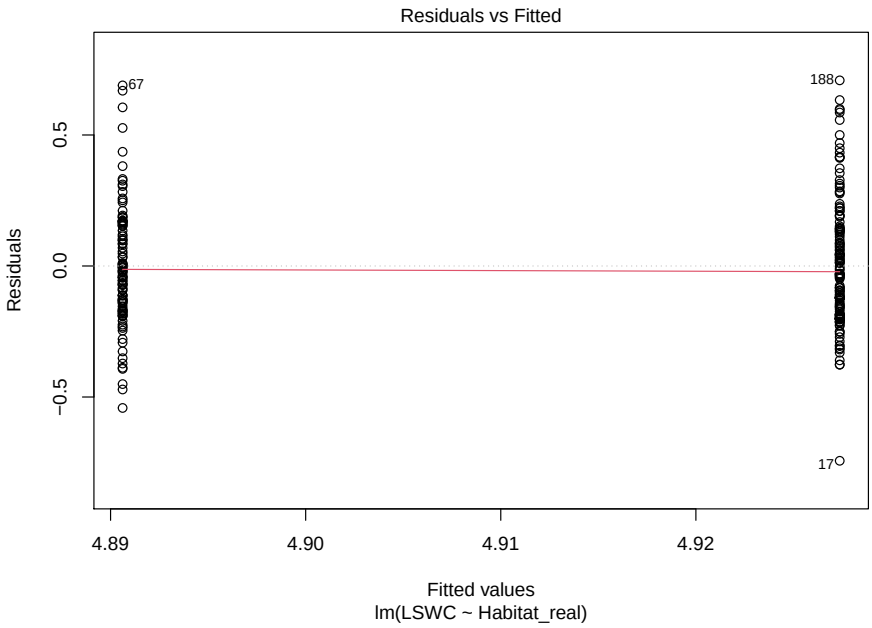
```
##
##  Bartlett test of homogeneity of variances
##
## data:  My_gmin.slope by Habitat_real
## Bartlett's K-squared = 1.6853787, df = 1, p-value = 0.1942113
```

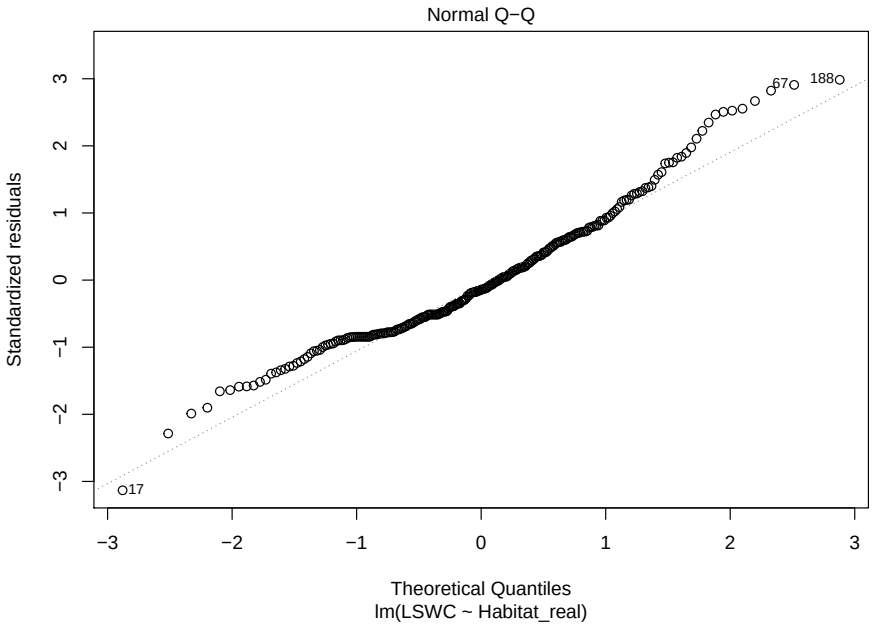
```
## # A tibble: 1 x 7
##   .y.               n statistic   DFn   DFd     p method
## * <chr>         <int>     <dbl> <dbl> <dbl> <dbl> <chr>
## 1 My_gmin.slope   265      3.38     1  234. 0.067 Welch ANOVA
```

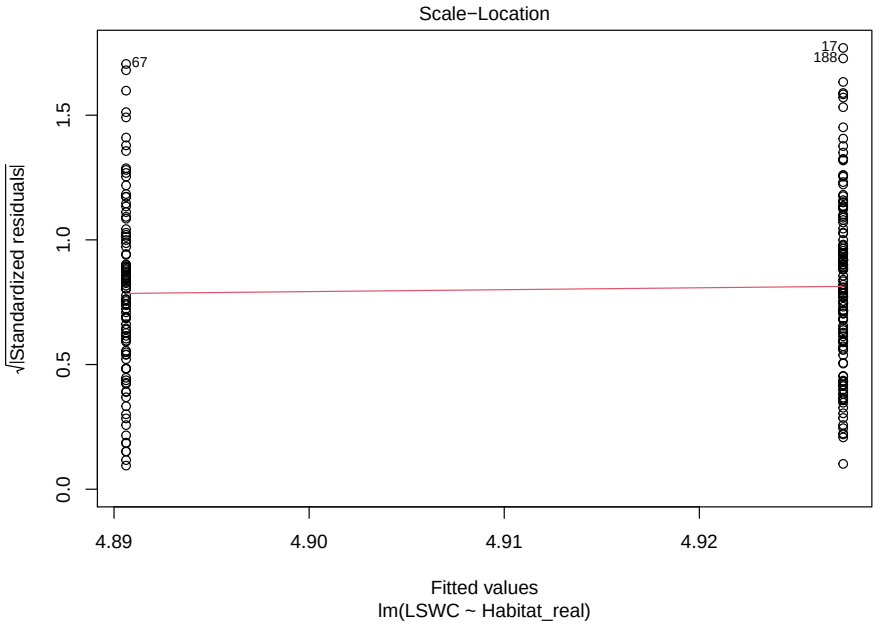
```
##
## Call:
```

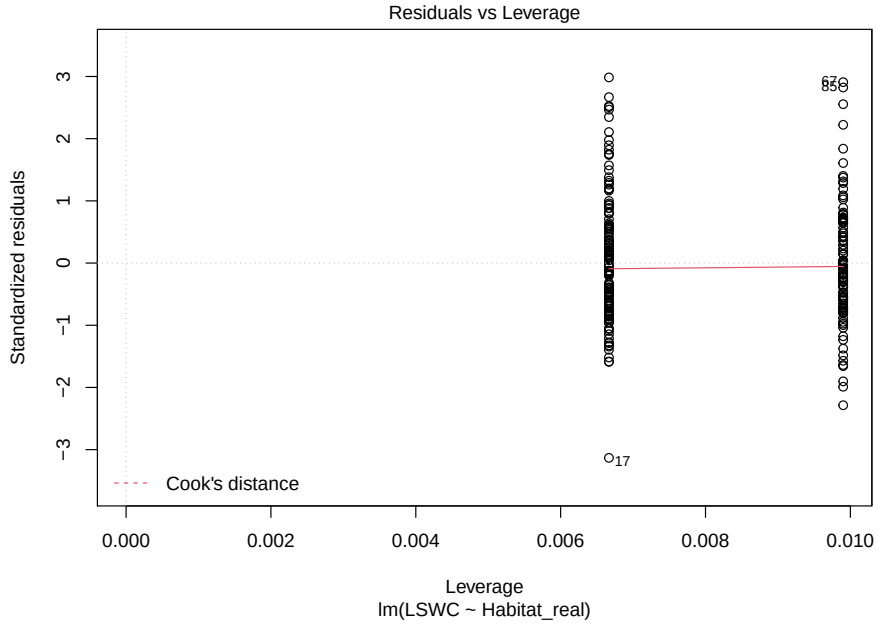
```
## lm(formula = LSWC ~ Habitat_real, data = data)
##
## Residuals:
##         Min          1Q      Median          3Q         Max
## -0.74338656 -0.17445207 -0.03452023  0.14166074  0.70845148
##
## Coefficients:
##                   Estimate  Std. Error   t value Pr(>|t|)
## (Intercept)     4.92737223  0.01944433 253.40920  < 2e-16 ***
## Habitat_realTF -0.03675476  0.03065273  -1.19907  0.23164
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2381434 on 249 degrees of freedom
##   (14 observations deleted due to missingness)
## Multiple R-squared:  0.005741019,    Adjusted R-squared:  0.001748011
## F-statistic: 1.437768 on 1 and 249 DF,  p-value: 0.2316407
```

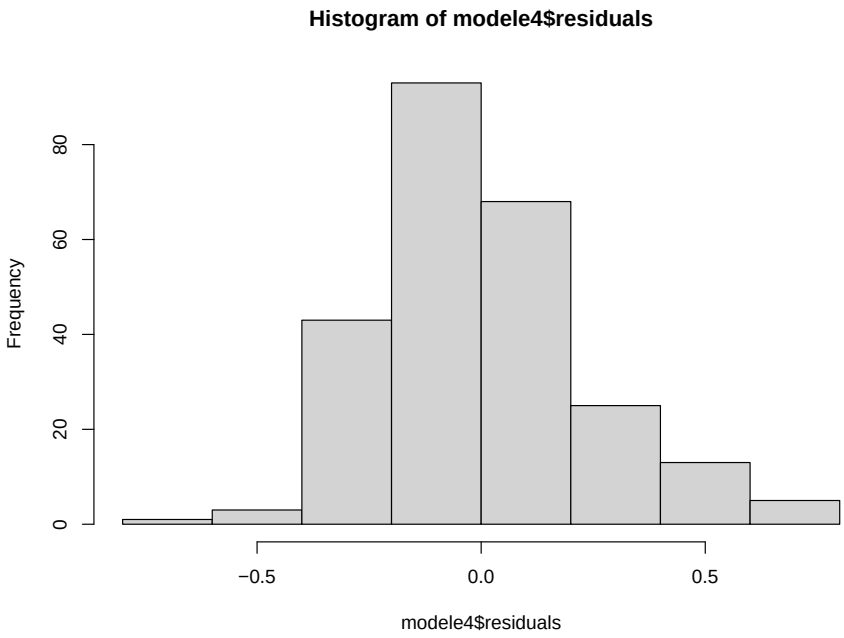
```
##
## Call:
## lm(formula = LSWC ~ Habitat_preference + Habitat_real, data = data)
##
## Residuals:
##         Min          1Q      Median          3Q         Max
## -0.66575519 -0.13759999 -0.04424304  0.12440242  0.68452761
##
## Coefficients:
##                                Estimate  Std. Error   t value   Pr(>|t|)
## (Intercept)                  5.01850472  0.02767850 181.31420 < 2.22e-16 ***
## Habitat_preferenceGeneralist -0.16876385  0.03766566  -4.48058 1.1392e-05 ***
## Habitat_preferenceTF         -0.18089317  0.06022211  -3.00377  0.0029408 **
## Habitat_realTF                0.04556021  0.03879687   1.17433  0.2413951
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2299149 on 247 degrees of freedom
##   (14 observations deleted due to missingness)
## Multiple R-squared:  0.0807069,  Adjusted R-squared:  0.0695414
## F-statistic: 7.228237 on 3 and 247 DF,  p-value: 0.0001143635
```

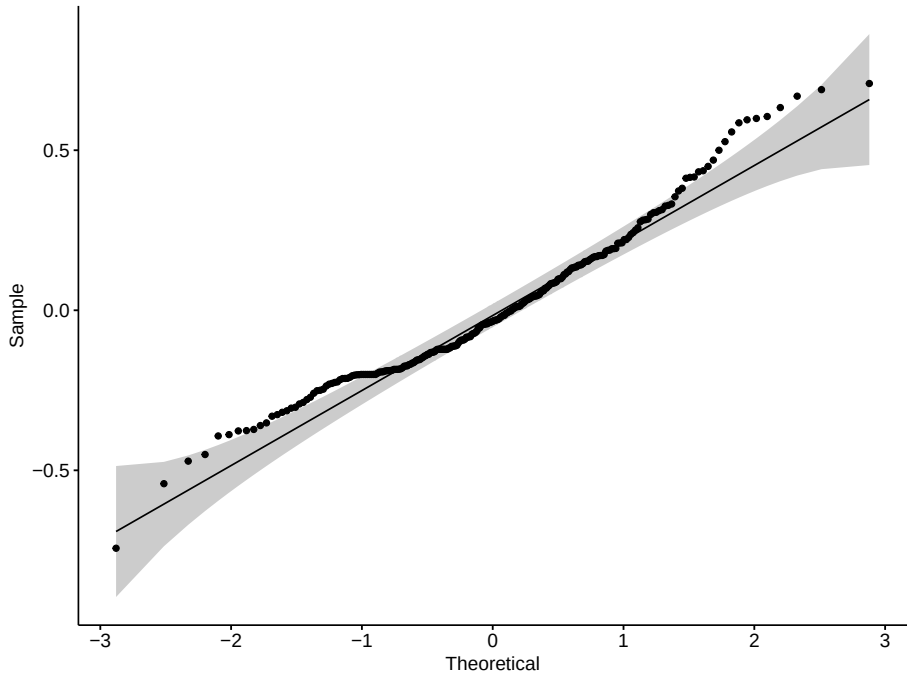

Residuals vs Fitted
Residuals
Fitted values
lm(LSWC ~ Habitat_real)
67
188
17
0.5
0.0
−0.5
4.89
4.90
4.91
4.92


Normal Q−Q
Standardized residuals
Theoretical Quantiles
lm(LSWC ~ Habitat_real)
67
188
17
3
2
1
0
−1
−2
−3


Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
67
17
188
4.89
4.90
4.91
4.92
Fitted values
lm(LSWC ~ Habitat_real)

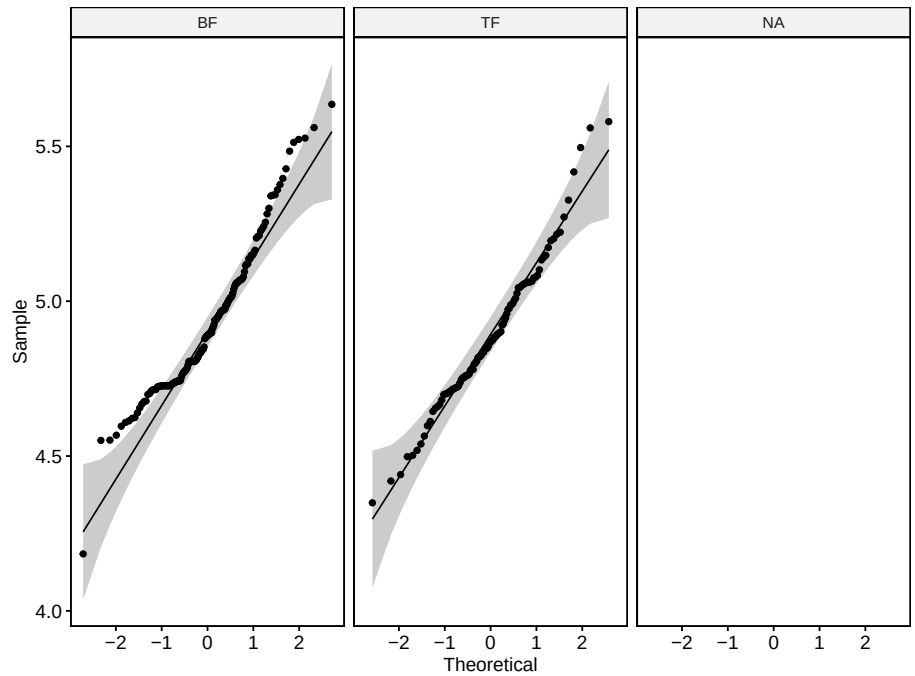

Residuals vs Leverage
Standardized residuals
−3
−2
−1
0
1
2
3
67
85
17
Cook's distance
0.000
0.002
0.004
0.006
0.008
0.010
Leverage
lm(LSWC ~ Habitat_real)


Histogram of modele4$residuals
Frequency
0
20
40
60
80
−0.5
0.0
0.5
modele4$residuals


Sample
0.5
0.0
−0.5
−3
−2
−1
0
1
2
3
Theoretical

```
## # A tibble: 1 x 3
```

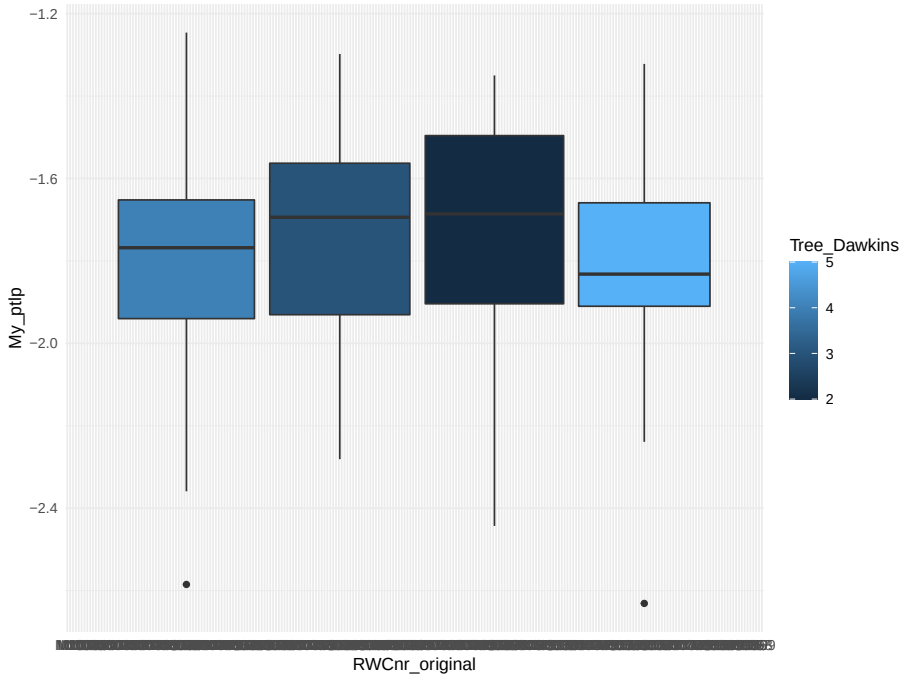
```
##   variable           statistic   p.value
##   <chr>                  <dbl>     <dbl>
## 1 residuals(modele4)     0.968 0.0000234
```




```
## # A tibble: 1 x 4
##     df1   df2 statistic     p
##   <int> <int>     <dbl> <dbl>
## 1     1   249     0.146 0.703
```

```
##
##  Bartlett test of homogeneity of variances
##
## data:  LSWC by Habitat_real
## Bartlett's K-squared = 0.092181246, df = 1, p-value = 0.7614223
```

```
## # A tibble: 1 x 7
##   .y.       n statistic   DFn   DFd     p method
## * <chr> <int>     <dbl> <dbl> <dbl> <dbl> <chr>
## 1 LSWC    265      1.45     1  219. 0.229 Welch ANOVA
```


−1.2
−1.6
−2.0
−2.4
My_ptlp
RWCnr_original
Tree_Dawkins
5
4
3
2

Chapter 11

Vein traits

The leaf is a critical component in the plant water transport system, accounting for 30% or more of whole-plant hydraulic resistance (Sack and Holbrook, 2006). Leaf venation architecture has numerous common functions across plant species—see Sack & Scoffoni, 2013 for review. Briefly, the leaf venation serves for mechanical support (Niklas, 1999), sugar and hormone transport in the phloem (Kehr & Buhtz, 2008), and, via the xylem, the replacement of water lost to transpiration when the stomata open for photosynthesis (Sack & Holbrook, 2006). However, venation architecture is highly diverse across species (Uhl & Mosbrugger, 1999; Roth-Nebelsick et al., 2001; Sack & Frole, 2006; Ellis et al., 2009; Brodribb et al., 2010). In dicotyledons, the leaf venation system typically consists of three orders of major veins and up to five higher orders of minor veins embedded in the mesophyll, with the vein orders arranged in a hierarchy; lower order veins are larger in diameter, with greater xylem conduit numbers and sizes, whereas higher order veins have greater length per leaf area (VLA; Sack & Holbrook, 2006; McKown et al., 2010). Total leaf VLA has been shown to correlate with maximum hydraulic conductance and photosynthetic rate per area across species (Sack & Frole, 2006; Brodribb et al., 2007) and tends to be higher for species growing in high light. Major VLA has been found to play a role in determining the damage tolerance of the vein system, and in leaf drought tolerance (Sack et al., 2008; Scoffoni et al., 2011).

11.1 Hypothesis

- Smaller leaves tend to have a higher major VLA. For SFF specialists, the replacement of water lost to transpiration during photosynthesis would be no problem during the wet season, since water is not a limiting factor. We collected the leaves during the dry season, therefore, SFF specialists

would either have an adapted mechanisms to overcome water scarcity or not.
- Leaves from the driest habitat, *Terra firme* are smaller and have a higher major VLA than the leaves from seasonally flooded habitats.
- Leaves from Bafog (lower precipitation regime) are smaller with a higher major VLA than leaves in Kaw (higher precipitation regime). Paracou leaves major VLA values would range in the middle.

Litterature :

Smaller leaves tend to have higher major VLA intrinsically as a result of their development. The drought tolerance and protection conferred by higher major VLA in smaller leaves provides an explanation for the greater abundance of smaller leaves in drier and more exposed habitats (Scoffoni et al., 2011).

Drought impacts less leaves with higher major vein density because they can provide more numerous water flow pathways around the damaged vein. (Scoffoni et al 2011)

Smaller leaves, which generally have higher major vein density, would thus have lower hydraulic vulnerability. Tests using simulations with a spatially explicit model confirmed that smaller leaves with higher major vein density were more tolerant of major vein embolism.(Scoffoni et al 2011)

From Loundry *et al* 2012:

- SFF : Leaves from SFF have a higher production rates. This result correlates with the higher soil resources found in these habitats and rapid rates of turnover of stems in these forests due to seasonal flooding (Ferry et al. 2010, Baraloto et al. 2011).
- no herbivory trends differing from SFF and TF
- but sapling stage not adult or seedling stage
- more herbivory found in Peru than in French Guyana

From Scoffoni *et al* 2011:

- Smaller leaves more frequent in dry habitats
- Smaller leaves have thinner boundary layer: more rapid convective cooling

- Direct hydraulic benefit of small leaves, if their greater major vein redundancy protects K leaf from decline and thus contributes to drought tolerance.

11.2 Materials and Method

Leaf area (LA, mm2) Major vein density (Major VLA, mm/mm2) : vein length per unit area Sum of vein densities for 1°, 2° and 3° veins (mm mm-2) Double

sided lamp flatbed scanner ImageJ

Why a higher VLA? To confer redundancy and drought tolerance

Measurements from Scoffoni's Lab. These measures were calculated by a student group project from January - March 2022.

11.3 Statistical analysis

- ANOVAs for the differences between SFF and TF habitat
- ANOVAs for the differences between the three sites
- ANOVAs for the differences between habitat specialists and generalists Species nested within those categories

Additional data to confront VLA with :

Linear regression analysis to performe to test the correlations between the other variables.

- Ptlp
  - hypotheses : Species of TF habitat would exhibit greater expression of leaf drought tolerance traits than species from SFF habitat i.e. small leaves with high VLA associated with more negative values for osmotic potential at at turgor loss point.
- gmin (same VLA leaf)
  - Species of TF habitat would exhibit greater expression of leaf drought tolerance traits than species from SFF habitat i.e. small leaves with high VLA associated with lower minimal cuticular conductance.
- LSWC
  - Species of TF habitat would exhibit greater expression of leaf drought tolerance traits than species from SFF habitat i.e. small leaves with high VLA associated with higher LSWC more water per unit dry weight, that would provide a larger buffering capacity when water supply becomes limiting (higher water storage)
- Stomatal density *waiting for data*
- leaf chemistry *waiting for data*
  - Palatability was positively correlated with N content, SLA and water content of leaves and negatively correlated with leaf C content and leaf C/N-ratio. (Schadler et al 2003)

Paracou data: recheck LA and gmin!

FTH data :

Both FTH and Paracou data combined:

Bafog data:

11.4 Results

To do : -merge vein data set with the rest of data of each site through the individual code

11.4.1 Vein data joing with other traits

11.4.1.1 Merge

11.4.2 Across species from all habitats

11.4.3 SFF vs TF habitats

11.4.4 Specialists SFF vs Specialists TF vs Generalists

11.4.5 Bafog vs Kaw vs Paracou

11.4.6 VLA against other variables

VLA and LA

VLA and Ptlp

Preliminary and wrong because left joining by species. I want to left join by Code. !!! I tried by I can't this way…

11.4.7 Herbivory?

Chapter 12

Other venation traits

Traits that gives us info on the construction cost.

VC : vein density x $\times$ the vein diameter VP : Vein projected area per leaf area = vein diameter x vein density VV : Vein volume per leaf area = vein density x x (diameter/2)^2

For each vein order!!!

VC : the vein cross-sectional surface area per leaf area was estimated by multiplying the vein density by $\times$the vein diameter.

VP : The vein-projected area per leaf area *(the proportion of leaf lamina obstructed by vein)* was determined by multiplying the vein diameter by vein density. + global value :0.00712 for 2° veins (Sack et al 2012)

VV: Vein volume per leaf area was determined by multiplying the vein density by $\times$ the square of half the diameter.

Chapter 13

Waterlogging litterature

- From **Abiotic Stress Physiology of Horticultural Crops** *edited by N.K. Srinivasa Rao, K.S. Shivashankara, R.H. Laxman*

Section 1.3 : one of the earliest responses to soil flooding is a reduction in stomatal conductance, inhibition of stomatal aperture. Low O2 level may also reduce hydraulic conductivity and csqtly decrease root permeability. This descrease in hydraulic permeability may b liked to aqaporins [...] limiting water uptake and this leading to internal water deficit. Oxygen deficiency generally induces a rapid reduction in the rate of photoS which is generally considered as the result of reduced stomatal aperture, decrease in leaf chloro content + early leaf senescence and a reduction in LA. Waterlogged soils reduce the translocation of photoS product from 'source' to 'sink' root. As a result, the maintenance of photoS acitivity and accumulation of soluble sugar in roots are clearly important adaptations to flooding.

Plant responses to waterlogging include low stomatal conductance, decrease in leaf, root, and shoot dev't, changes in osmotic potential and nu uptake, reduced chloro content and photoS.

The oxygen diffusion is 10,000 times slower in waterlogged soils as compared to aerated soil.

The decrease in O evel in soils affects the bioavailability of nu and as well as the ability of root systems to uptake and transport water + mineral nu. Waterlogging also causes inhibition of N-uptake from the soil and reducedleaf concentrations of N, P, K, Ca, Mg in avocado + pea.

Chapter 14

general potential hypotheses

with waterlogging: i should find a decrease of stomatal conductance corrolated with an increase of leaf water potential.

Part IV

Part four : Notes

Chapter 15

26/01/2022

Metradica meeting

Presents:

- Ghislain
- Clément

- Sabrina
- Damien

- Jeanne
- Marion

Objectives:

1- les données 13C
2- les modèles joints

Notes:

(1)

L'estimation de l'efficience d'utilisation de l'eau intrinsèque peut être obtenue à l'aide de msures directes des échanges gazeux entre feuille et atm. Farquar 1982 propose un modèle simple reliant WUE (water use efficiency) à la discrimination isotopique du C au cours de la photosynthèse.

Relation WUE et δ 13C positive.

Drivers de la compo iso:

- Lumière (feuille d'ombre valeurs 13C nettement plus négatives)
- génétique
- sécheresse atm + sol
- VPD
- teneur en mx du sol
- ontogénie de l'individu
- conductance hydraulique

Attention: la compo iso n'est pas un trait lié à la sécheresse. Il caractérise *l'acclimation* à la sécheresse. Il est utile pour une expé en serre de comparer le δ 13C entre le groupe controle et le groupe dry. Une espèce sensible: on trouvera une forte différence vs. une espèce peu sensible, une faible différence. Et le *delta* ser aun trait de réponse à la sécheresse.

Travaux de Damien:

- 16 parcelles à travers la Guyane
- 9-10 sites

- ~100 arbres / parcelles
- ~1600 valeurs de compo iso de feuille d'arbres dominant
- 70 espèces différentes avec qq indet. par site; total de 392 espèces identifiées
- Moyenne valeur δ 13C ~ -30 %0 pour les forêts en Guyane
- Lorsque le climat est le même, on observe le même fonctionnement

(2)

Avant nous procédions par espèces, le but des modèles joints: faire avec plusieurs espèces en même temps.Les modèles joints de distribution des espèces permettent d'estimer la niche des espèces, de prédire leur distribution, tout en prenant en compte les intéractions entre espèces. Ils peuvent être utilisés pour déterminer les communautés ou assemblages d'espèces et comment ces assemblages changent spatialement, selon des gradients environnementaux.

Prendre en compte: la co-occurence des espèces + relation trait-env't pour déterminer la présence d'une espèce sur 1 site.

Si l'on inclu les données 13C dans les modèles joints: on s'attend à une abscence de réponse.

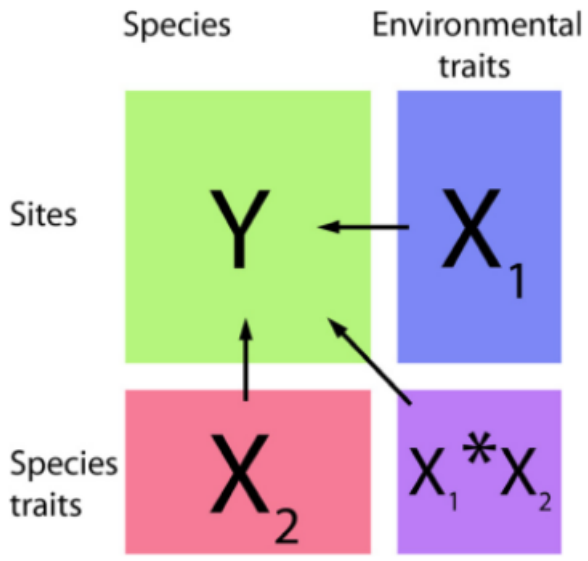


Figure 15.1: Modèle joint